Distinct, testable explanations across the three tests

Coverage and assay

All five models and all three tests are covered. 235 native replays and diagnostic controls passed spatial audit. The comparison measures the shortest tested pulse that persists after 10,000 units of recovery. 16 of 49 selected successful replay pulses end before half the old-pole contrast excess disappears. Pulse duration is a commitment dose and need not equal complete reorientation time.

1. Expanding domains versus spatial cue coverage

Wave-pinning and Holmes keep nearly constant plateau height while the initial coherent domain expands with L. A fixed support covers a decreasing fraction; relative support rescues many long-cell failures without changing peak amplitude. Prediction: varying support at fixed peak should preferentially change large-cell commitment. This geometric burden does not by itself establish constant edge speed or a linear time law.

2. Substrate starvation around a concentrated incumbent pole

Goryachev instead retains a narrow pole while its peak rises from about 6 at L=3 to 117 at L=39. Opposite-end inactive abundance falls from about 0.879 to 0.088. The cue acts through that substrate, so a constant patch peak supplies about tenfold less initial activation in the long cell. Prediction: changing substrate availability or the total-material preparation should alter this component; the present assay retains its exact original states.

3. Stronger gradient initiation versus redistribution/competition

At fixed physical slope the peak grows with L. In Wave-pinning and Holmes, matched-reference-peak ramps fail at native long-cell doses, while increased inactive mobility yields stronger recovered polarity. The longest fixed-width Wave-pinning case is rescued by mobility at its unchanged maximum pulse. These controls support distinct input-strength and transport/competition contributions, rather than a single erosion clock.

Two distinct mechanical mechanisms

4. Spatial adhesion/mechanical memory

deBelly requires rewriting an additional state: cue-only MCA freezing defeats all twelve sampled native-successful orientation doses across the three tests. Fast Rac response alone therefore does not commit those reversals. Unbound-MCA mobility changes can erase polarization itself. Prediction: measure MCA and mechanics at cutoff alongside Rac, and test retention under adhesion perturbation.

5. Global extension feedback gates commitment

Spring uses extension-dependent global deactivation and dilution, rather than local MCA memory. Removing extension feedback during the cue defeats all four sampled gradient doses, while some longer patch doses remain successful. Matching the gradient peak also defeats the longer native doses. Prediction: perturbing extension/restoring feedback should affect commitment differently for localized patches and whole-cell ramps.

Evidence is model- and test-specific

The following fifteen sections each give an interpretation, evidence and testable prediction. A mechanism supported in one geometry is not automatically assigned to another. Coupled Rac/Rho domains, local adhesion memory and a global extension gate are different state variables and require different measurements.

Boundary sensitivity is visible

Longest-length remeshing preserves 14 of 15 orientation classifications and 15 strength classifications at the native selected orientation doses or censor caps. deBelly fixed-width orientation changes. These are fixed-dose checks, not converged thresholds or mesh tests at every original-strength endpoint. Native scalar gates and full-field steady-state claims remain separate.

Tests of length mechanisms and practical limits

The suggested constant-speed erosion mechanism

It predicts an old-domain width that grows with L, similar physical edge speeds at different lengths, and a removal time explained by width/speed. The report measures all three independently. It must not be inferred solely from a positive duration slope. Narrow poles that increase peak height, pulses ending before removal, and major mobility effects require a different account.

Separate the stimulus from the internal mechanism

A constant-slope ramp has a peak proportional to length. The matched-peak control deliberately changes the slope only as a diagnostic, outside the native three-test comparison. A reversal of the activation trend under that control supports input-strength acceleration. Fixed and relative patch amplitudes never scale with length.

Shape/readout effects can condition apparent length trends

The strength gate uses a normalized first moment, whereas pole width, peak height and total activity measure shape and material. Adjacent passing/failing native pulses and both scalar thresholds are retained. A minimum in the curve can combine faster initiation at larger L with slower cleanup, mechanical rewriting or selective scalar stability. Recovery time is not added to the plotted pulse duration.

What fixed-dose controls establish

Interventions test trajectories at the original selected pulse, not a fully refined counterfactual threshold. Mobility changes can alter an equilibrium profile and the 1% strength readout as well as transport. A 401-point remeshing diagnostic checks the longest selected cell; it interpolates rather than independently re-establishing the start. The report preserves outcome changes instead of treating diagnostic completion as robustness.

Measurements and evidence

Pole amount integrates positive excess above the contemporaneous median in the central 5% of the cell, separately in each half. This removes broadly elevated activity under gradients. The boundary-connected domain threshold is that background plus half the initial pole contrast. Front fits use the first 80%-to-20% erosion segment, need four samples, and enter speed ratios only with R-squared at least 0.8. First crossings are sampled; absent crossings remain explicit. Raw fits and all quantities are in the CSV.

Wave-pinning · Fixed width 3

Expanding domains and competition for a conserved pool

At larger lengths the old active domain contains more material, while the initial inactive concentration stays nearly unchanged. The cue first creates a competitor; transport and competition after withdrawal determine whether it persists. Mobility can change commitment without changing the original cue. The fixed support covers a decreasing fraction of a longer cell. The L=39 case is censored natively but passes after fourfold inactive mobility at the same maximum tested pulse; a single remeshing does not rescue it.

Measured evidence

Old-width long/short=7.379; old-excess amount=10.79; opposite-end inactive ratio=0.9956; initial added activation-source ratio=0.9934. Control failures at native doses: mesh_401 1/1, mobility_4x 0/4.

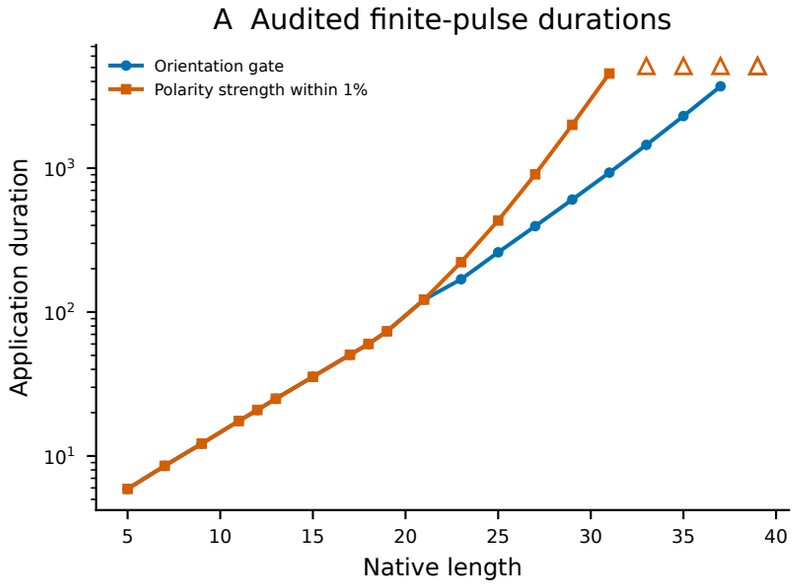
Testable prediction

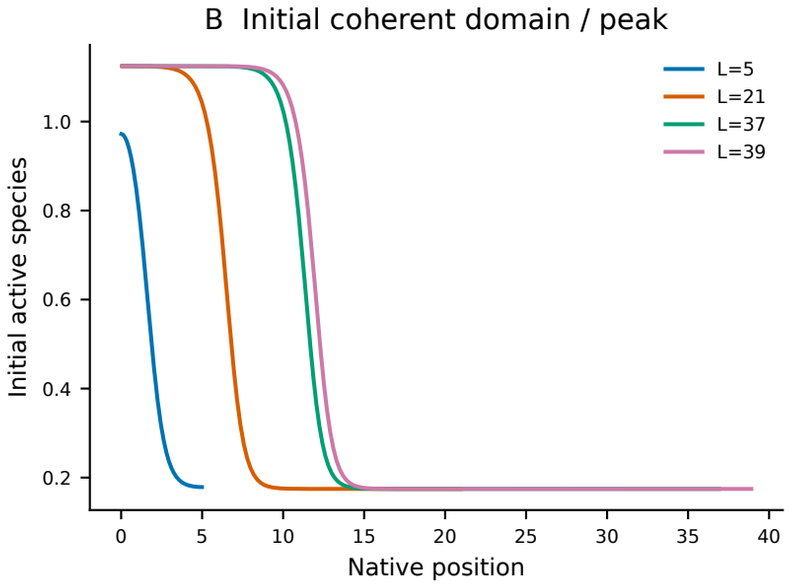
Measure right activation, old-domain retreat and inactive gradients separately. Increasing inactive mobility should change long-cell cleanup or commitment more than the initial local cue source.

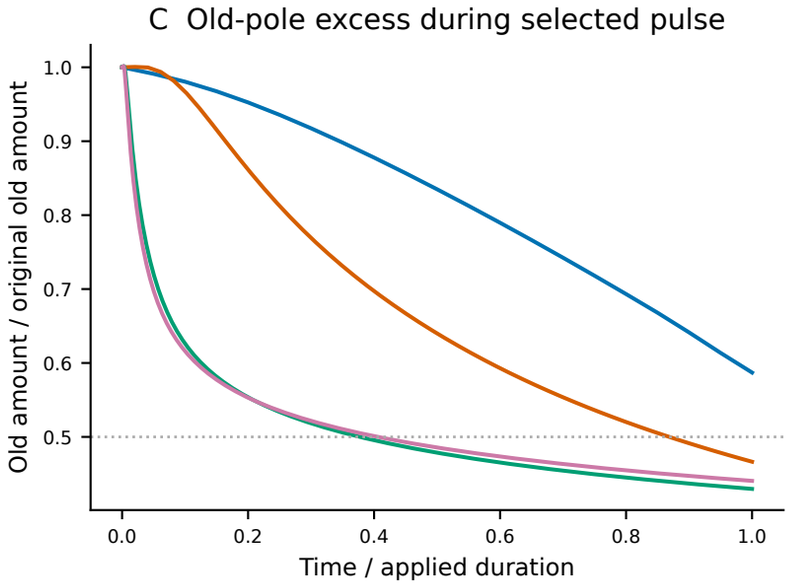
Scope of the inference

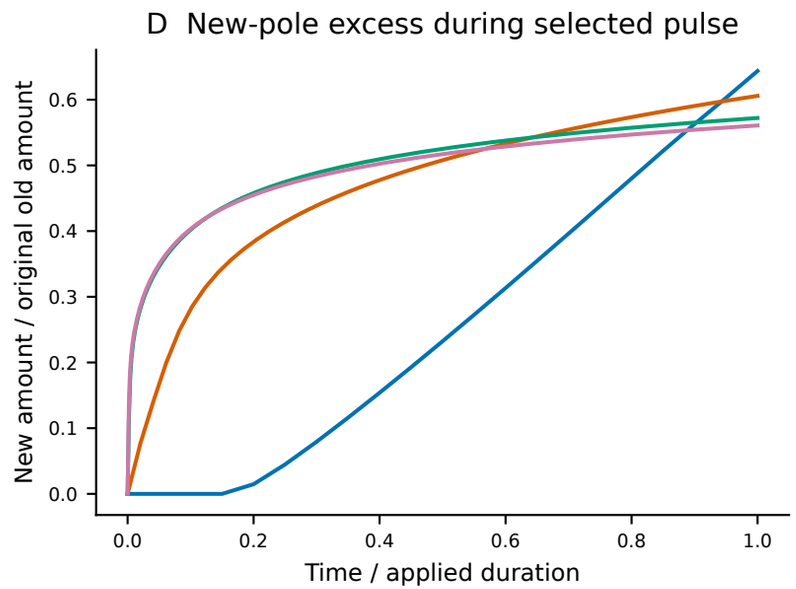
Trajectory and fixed-dose evidence in this native regime. Counterfactual thresholds were not fully searched; frozen-scale field flags and mesh sensitivity remain separate from scalar success.

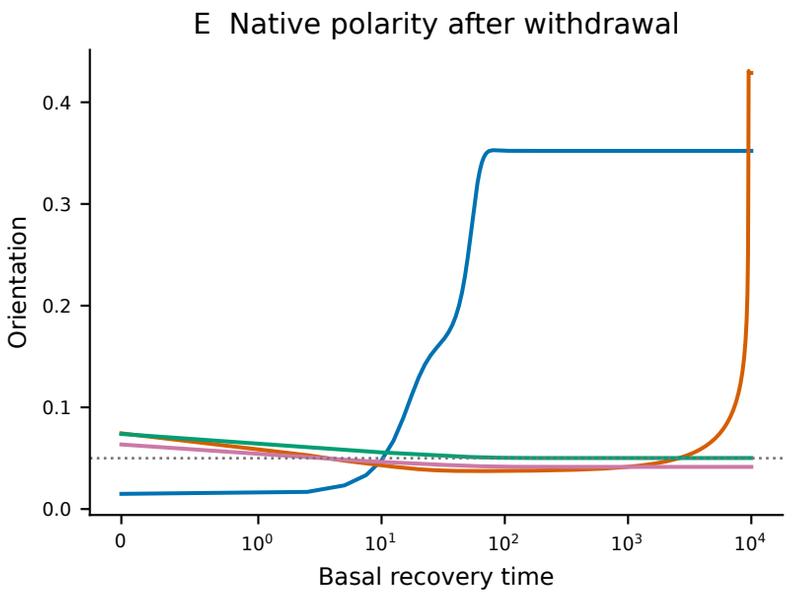
Wave-pinning · Fixed width 3

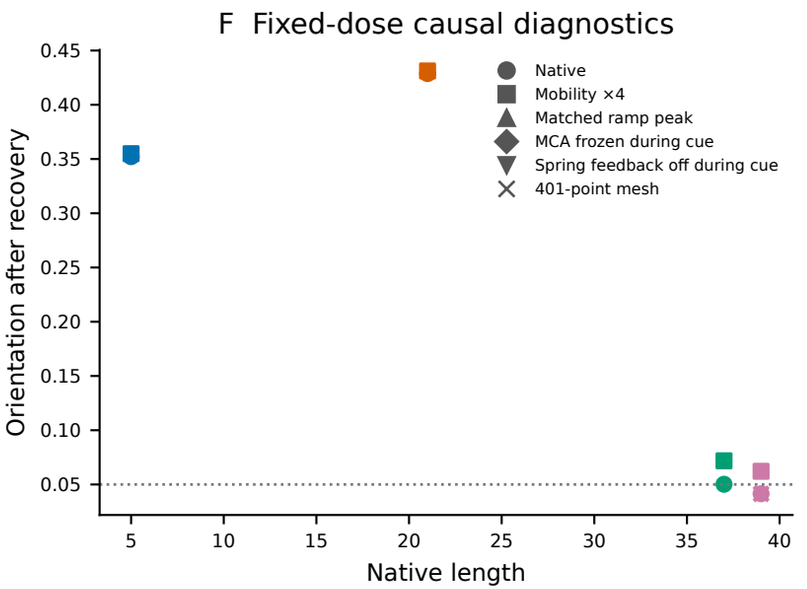




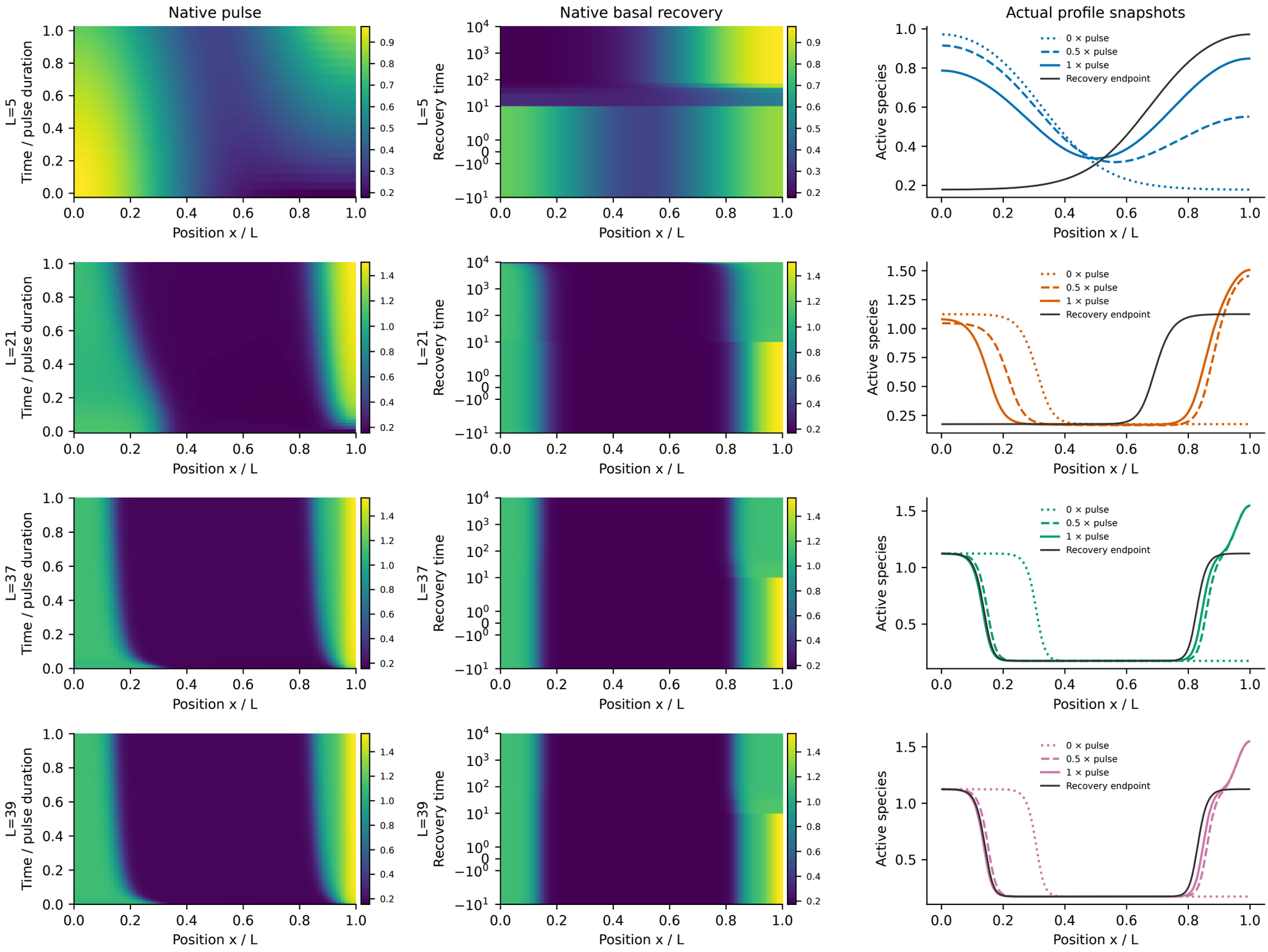








Amounts use excess above the current mid-cell median in each half; domain edges use that background plus half the initial pole contrast. Controls are fixed-dose diagnostics.



Native scalar-selected pulses. Separate panel color scales preserve concentration changes; these are not independently verified steady states.

Wave-pinning · Relative width 20%

Expanding domains and competition for a conserved pool

At larger lengths the old active domain contains more material, while the initial inactive concentration stays nearly unchanged. The cue first creates a competitor; transport and competition after withdrawal determine whether it persists. Mobility can change commitment without changing the original cue. The cue keeps the same footprint fraction and peak as length grows. All eligible lengths pass orientation, but the longer-cell full-strength threshold and field stability are distinct. Mobility improves the longest sampled orientation endpoint.

Measured evidence

Old-width long/short=7.776; old-excess amount=11.37; opposite-end inactive ratio=0.9956; initial added activation-source ratio=0.9956. Control failures at native doses: mesh_401 0/1, mobility_4x 0/3.

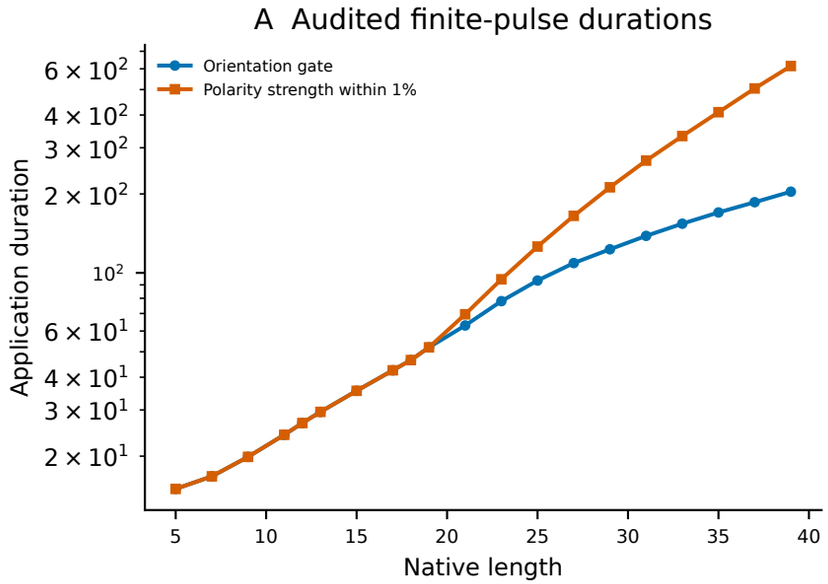
Testable prediction

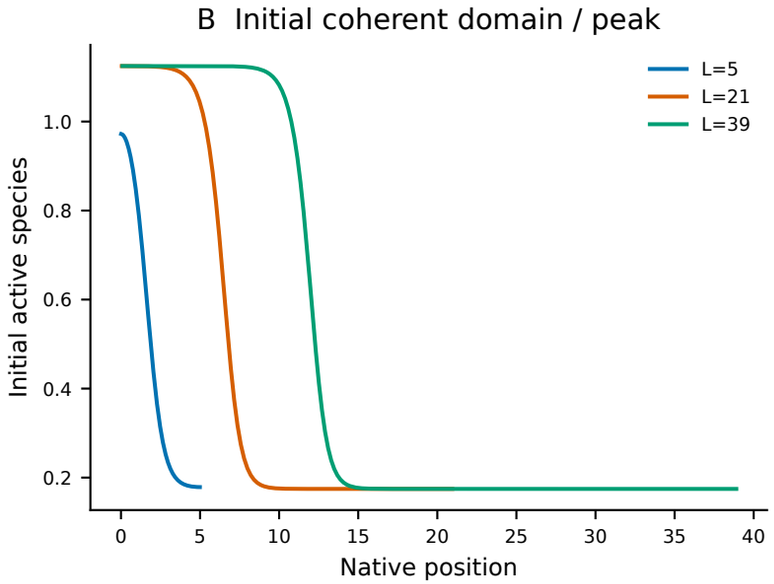
Measure right activation, old-domain retreat and inactive gradients separately. Increasing inactive mobility should change long-cell cleanup or commitment more than the initial local cue source.

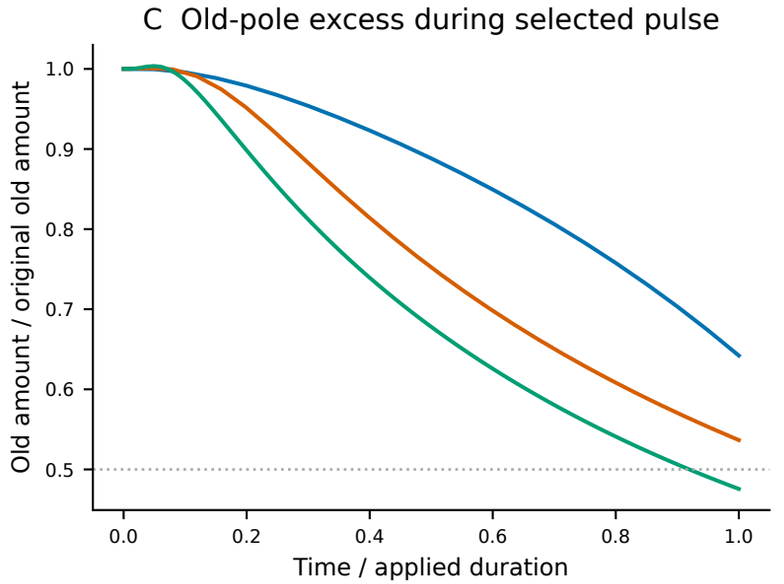
Scope of the inference

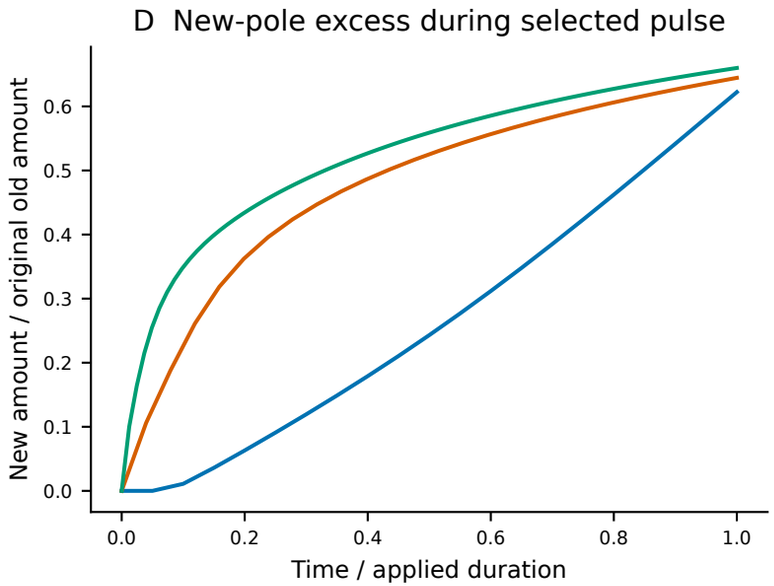
Trajectory and fixed-dose evidence in this native regime. Counterfactual thresholds were not fully searched; frozen-scale field flags and mesh sensitivity remain separate from scalar success.

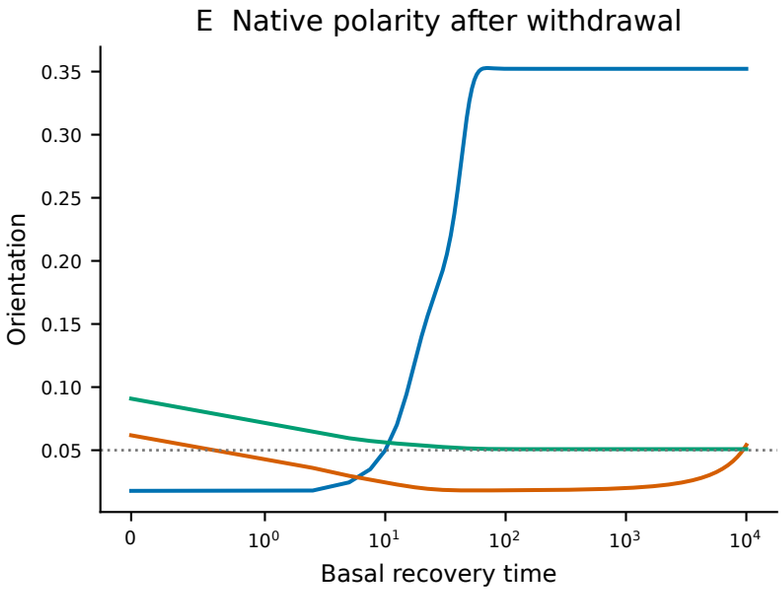
Wave-pinning · Relative width 20%

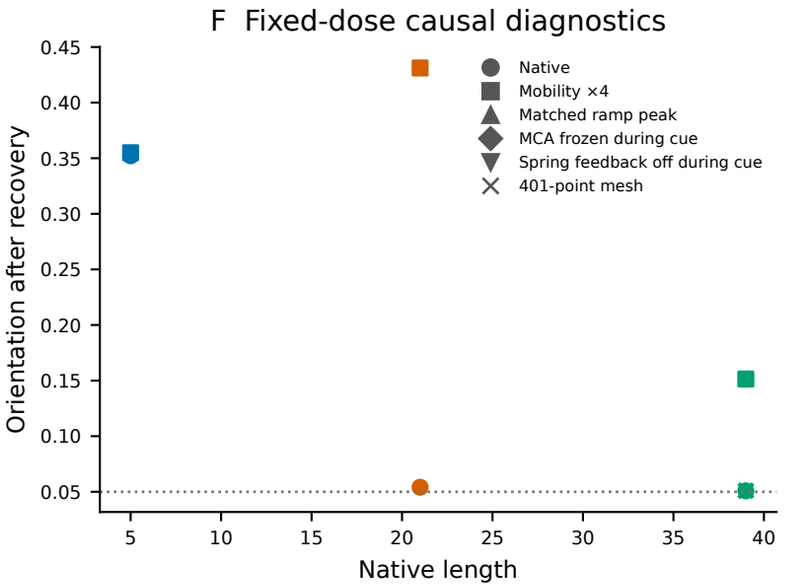




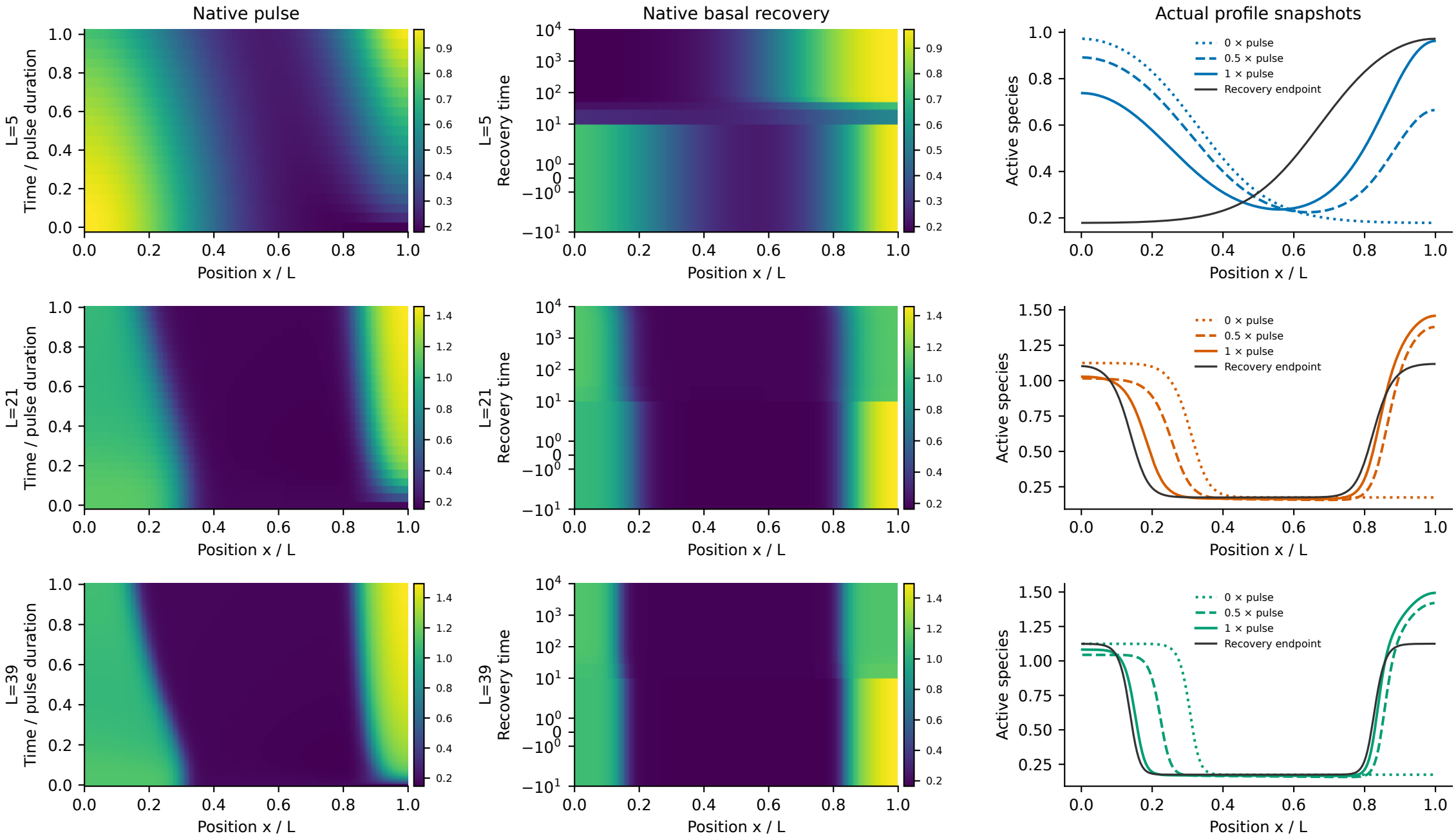








Amounts use excess above the current mid-cell median in each half; domain edges use that background plus half the initial pole contrast. Controls are fixed-dose diagnostics.



Native scalar-selected pulses. Separate panel color scales preserve concentration changes; these are not independently verified steady states.

Wave-pinning · Constant physical slope

Expanding domains and competition for a conserved pool

At larger lengths the old active domain contains more material, while the initial inactive concentration stays nearly unchanged. The cue first creates a competitor; transport and competition after withdrawal determine whether it persists. Mobility can change commitment without changing the original cue. The ramp peak increases with length because its physical slope is fixed. A matched-reference-peak ramp fails at the native long-cell doses, while fourfold inactive mobility preserves reversal and yields stronger recovered polarity. This supports opposing input-strength and pool/domain-cleanup contributions to the nonmonotonic curve.

Measured evidence

Old-width long/short=7.776; old-excess amount=11.37; opposite-end inactive ratio=0.9956; initial added activation-source ratio=7.766. Control failures at native doses: matched_gradient_peak 2/4, mesh_401 0/1, mobility_4x 0/4.

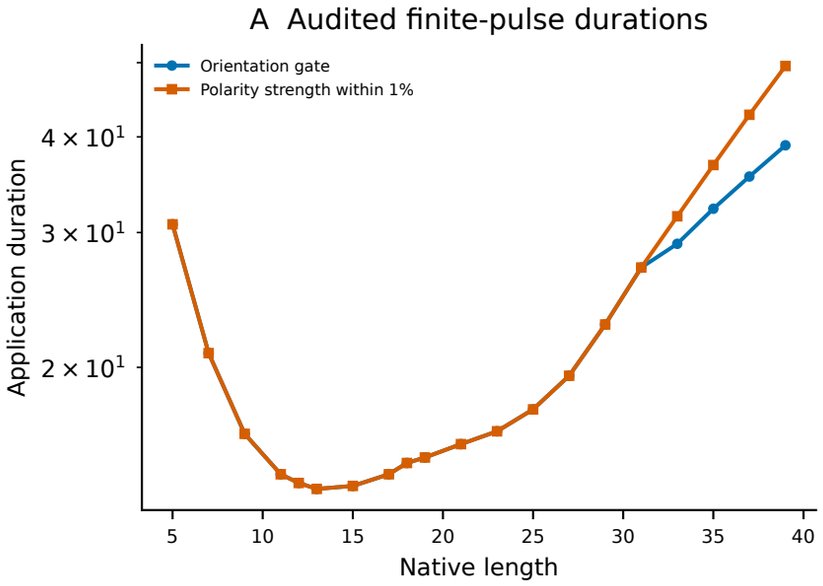
Testable prediction

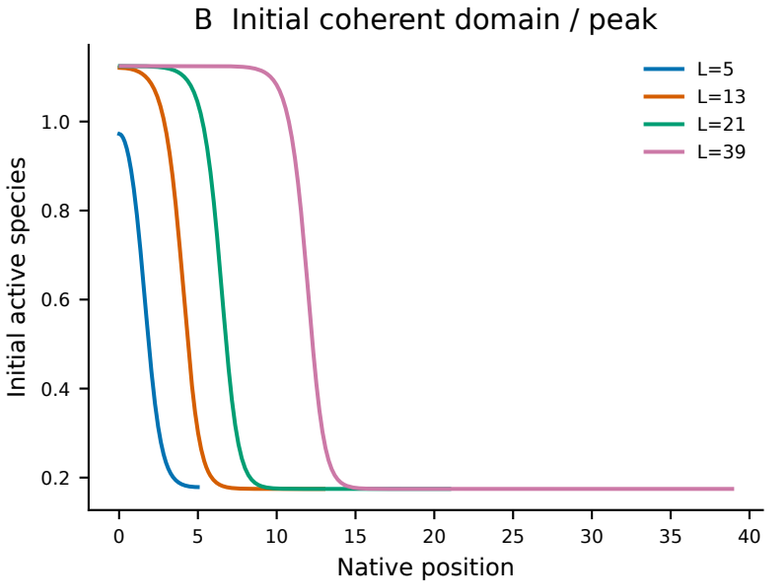
Measure right activation, old-domain retreat and inactive gradients separately. Increasing inactive mobility should change long-cell cleanup or commitment more than the initial local cue source.

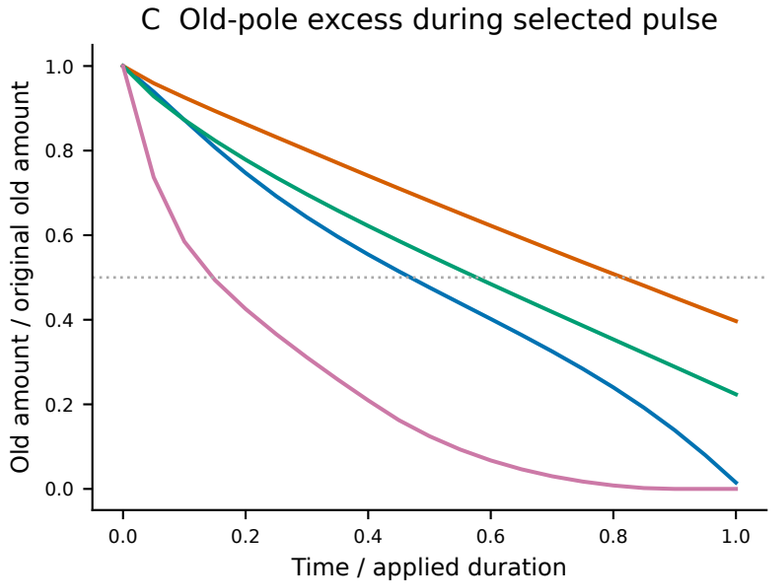
Scope of the inference

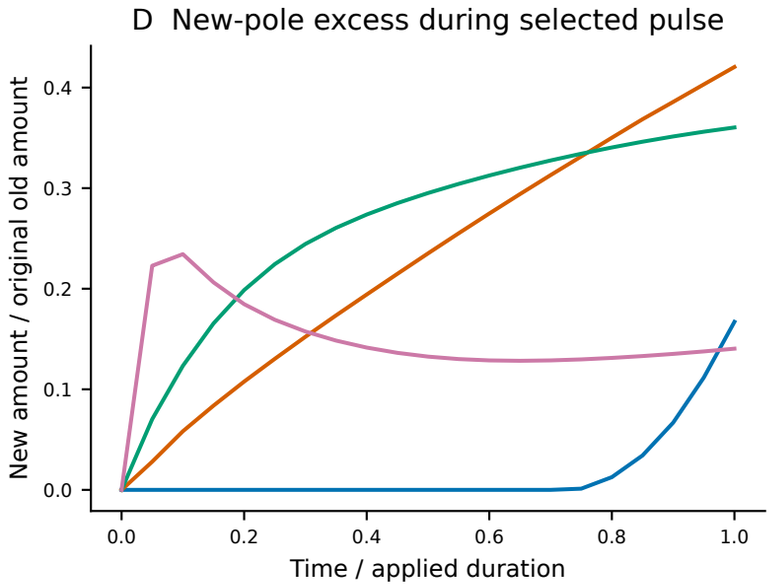
Trajectory and fixed-dose evidence in this native regime. Counterfactual thresholds were not fully searched; frozen-scale field flags and mesh sensitivity remain separate from scalar success.

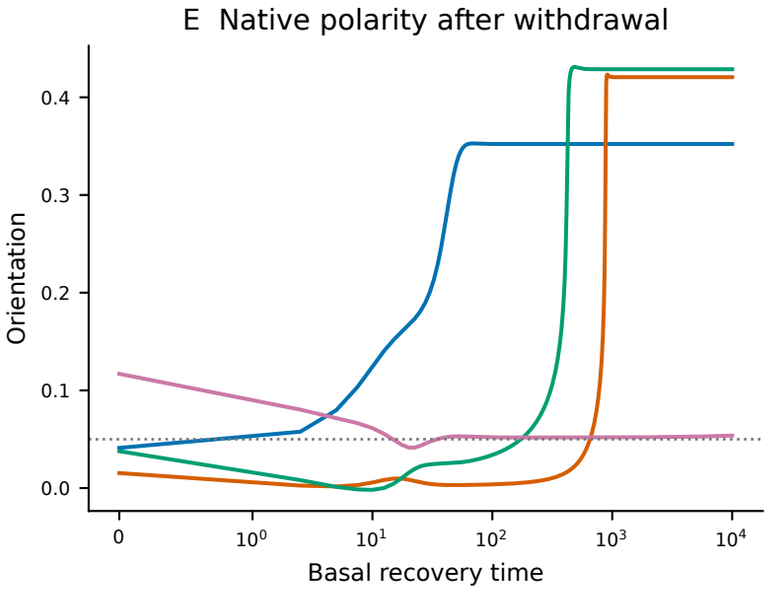
Wave-pinning · Constant physical slope

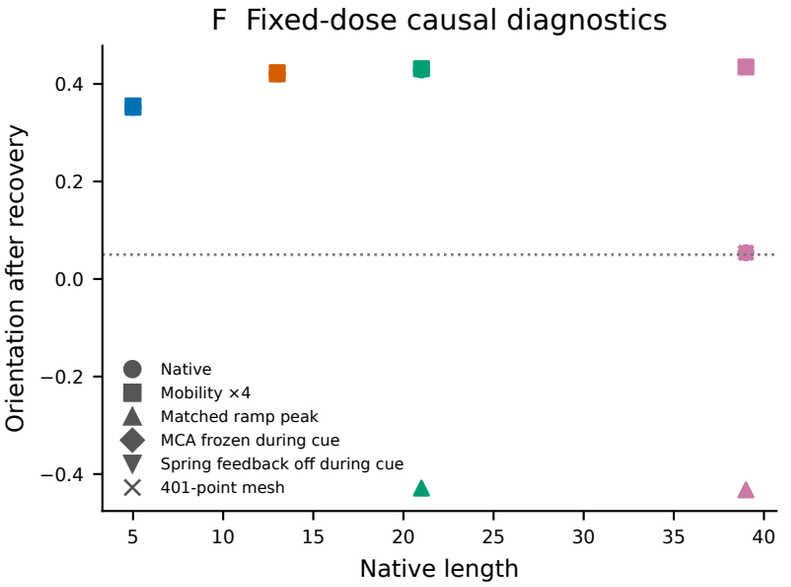






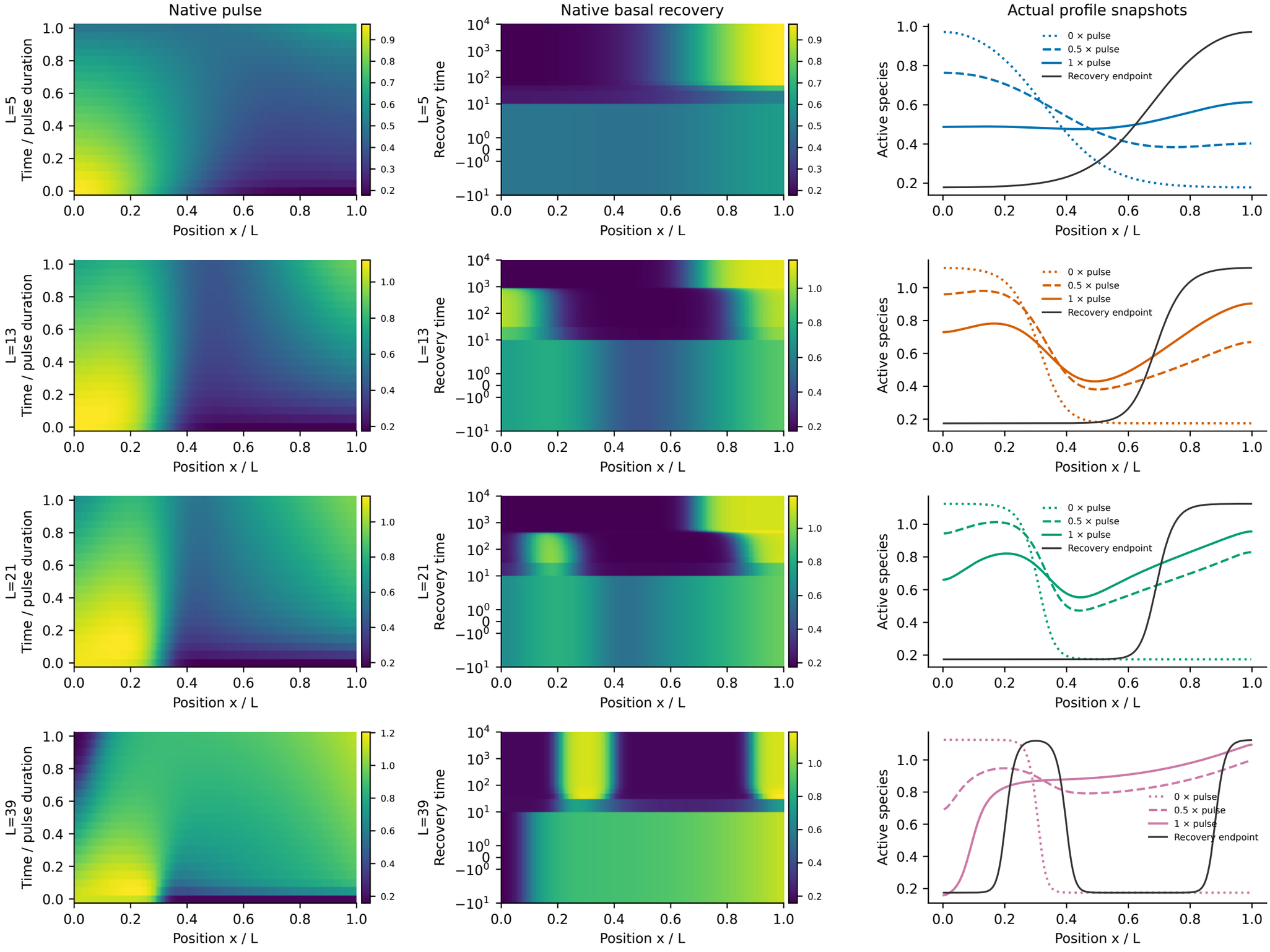






Amounts use excess above the current mid-cell median in each half; domain edges use that background plus half the initial pole contrast. Controls are fixed-dose diagnostics.

Wave-pinning · Constant physical slope · selected native spatial histories



Native scalar-selected pulses. Separate panel color scales preserve concentration changes; these are not independently verified steady states.

Goryachev · Fixed width 3

Mass sequestration in a concentrated self-amplifying pole

The incumbent pole stays about the same physical width but becomes much taller with length. Its inactive substrate at the opposite end falls sharply. The native added activation term is cue times inactive concentration, so equal patch peaks do not produce equal biochemical forcing. Slow competition and reconcentration of the conserved pool remain after cue initiation. The fixed support covers a decreasing fraction of a longer cell. The fixed K=64 amplitude rescues all eligible lengths, while the diminishing substrate still reduces initial added activation with length.

Measured evidence

Old-width long/short=1.099; old-excess amount=27.05; opposite-end inactive ratio=0.0998; initial added activation-source ratio=0.09955. Control failures at native doses: mesh_401 0/1, mobility_4x 0/3.

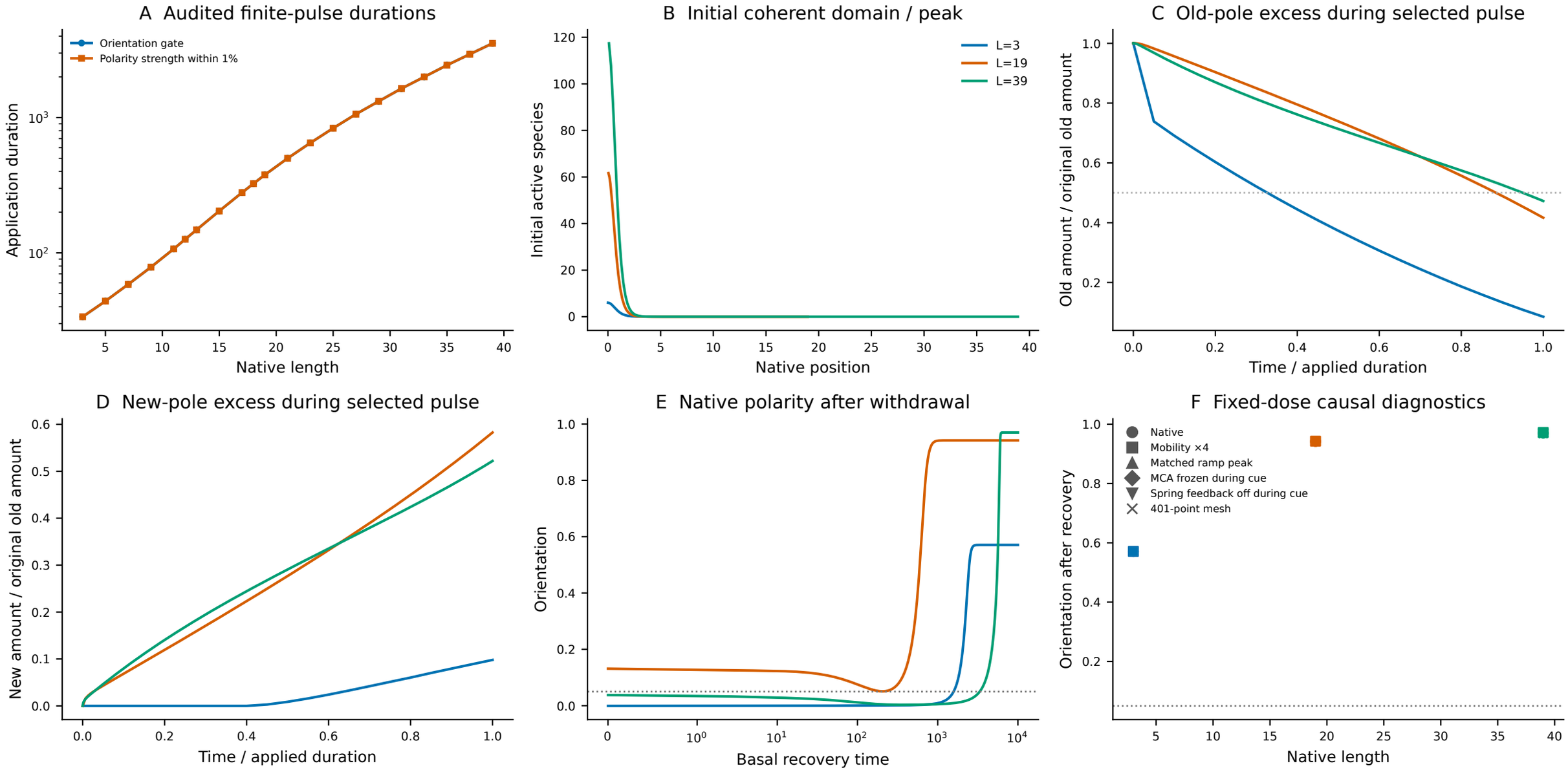
Testable prediction

Measure old-pole height and opposite-end inactive abundance, rather than assuming a widening domain. Restoring substrate availability or using a separate fixed-total-material preparation should reduce the length dependence of initial patch activation; this proposed test is not part of the native comparison.

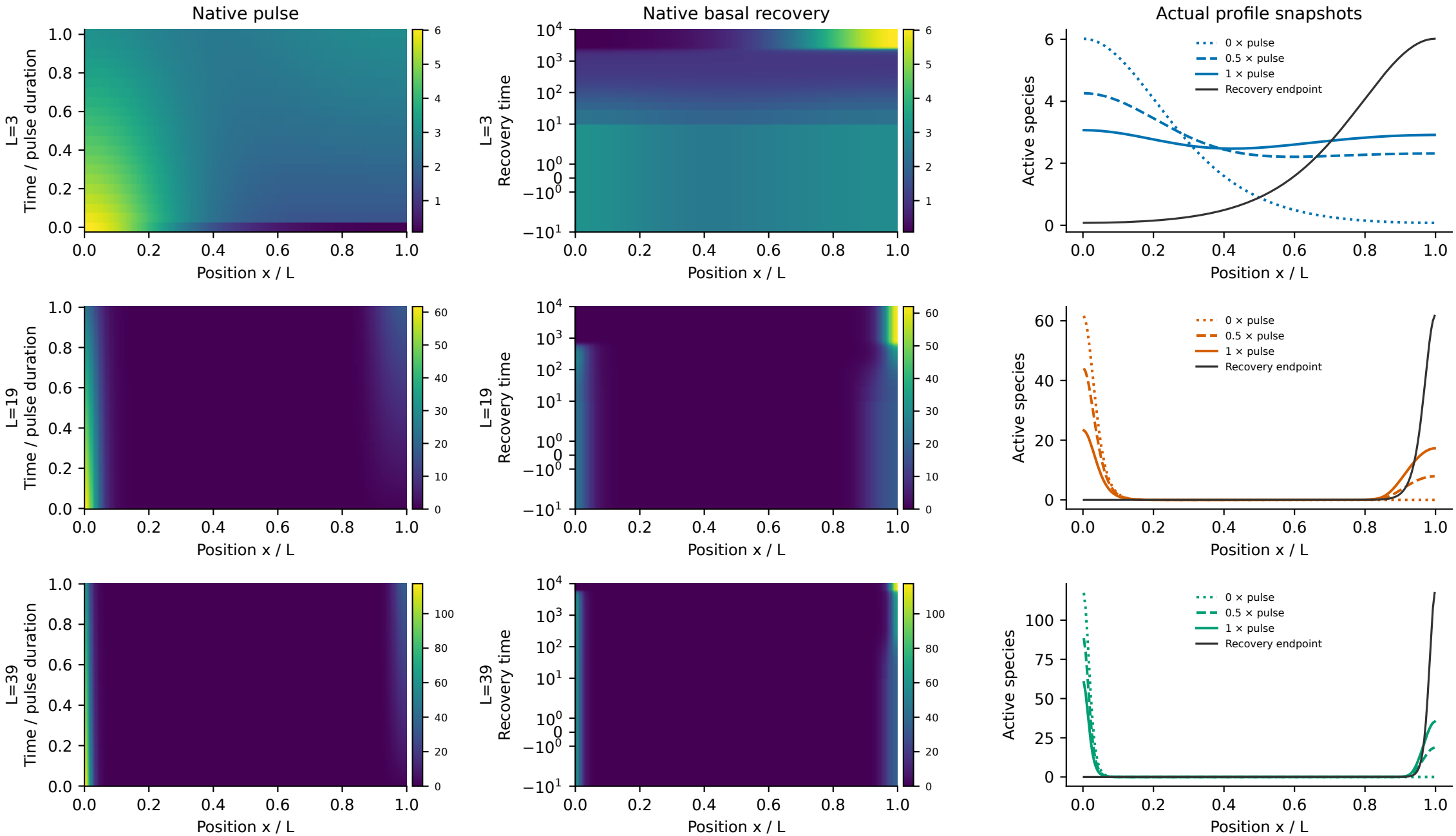
Scope of the inference

Trajectory and fixed-dose evidence in this native regime. Counterfactual thresholds were not fully searched; frozen-scale field flags and mesh sensitivity remain separate from scalar success.

Goryachev · Fixed width 3



Amounts use excess above the current mid-cell median in each half; domain edges use that background plus half the initial pole contrast. Controls are fixed-dose diagnostics.



Native scalar-selected pulses. Separate panel color scales preserve concentration changes; these are not independently verified steady states.

Goryachev · Relative width 20%

Mass sequestration in a concentrated self-amplifying pole

The incumbent pole stays about the same physical width but becomes much taller with length. Its inactive substrate at the opposite end falls sharply. The native added activation term is cue times inactive concentration, so equal patch peaks do not produce equal biochemical forcing. Slow competition and reconcentration of the conserved pool remain after cue initiation. The cue keeps the same footprint fraction and peak as length grows. All eligible lengths reverse at the fixed larger peak. Greater support helps without changing the strongly concentrated initial-pole regime.

Measured evidence

Old-width long/short=1.099; old-excess amount=27.05; opposite-end inactive ratio=0.0998; initial added activation-source ratio=0.0998. Control failures at native doses: mesh_401 0/1, mobility_4x 0/3.

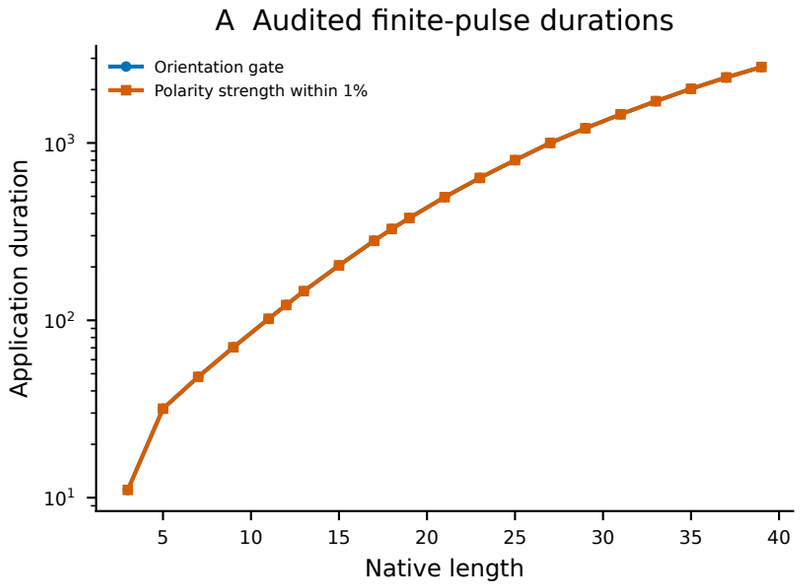
Testable prediction

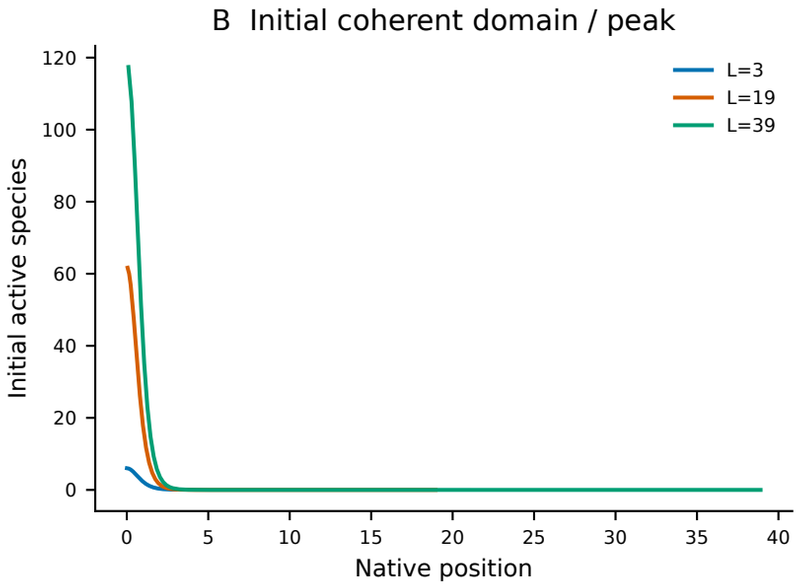
Measure old-pole height and opposite-end inactive abundance, rather than assuming a widening domain. Restoring substrate availability or using a separate fixed-total-material preparation should reduce the length dependence of initial patch activation; this proposed test is not part of the native comparison.

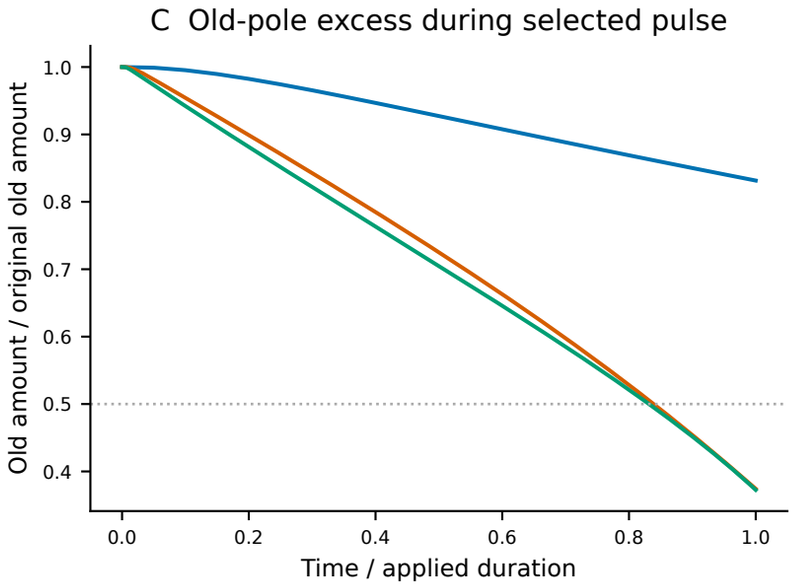
Scope of the inference

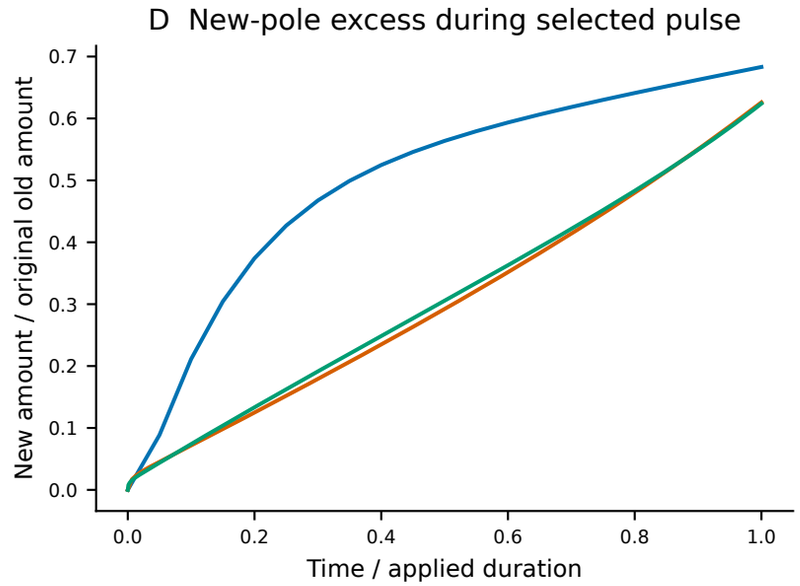
Trajectory and fixed-dose evidence in this native regime. Counterfactual thresholds were not fully searched; frozen-scale field flags and mesh sensitivity remain separate from scalar success.

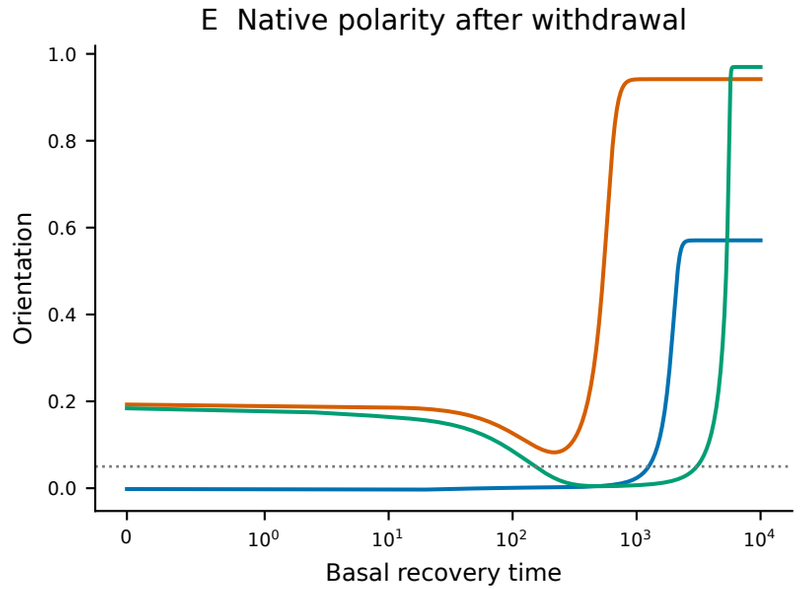
Goryachev · Relative width 20%

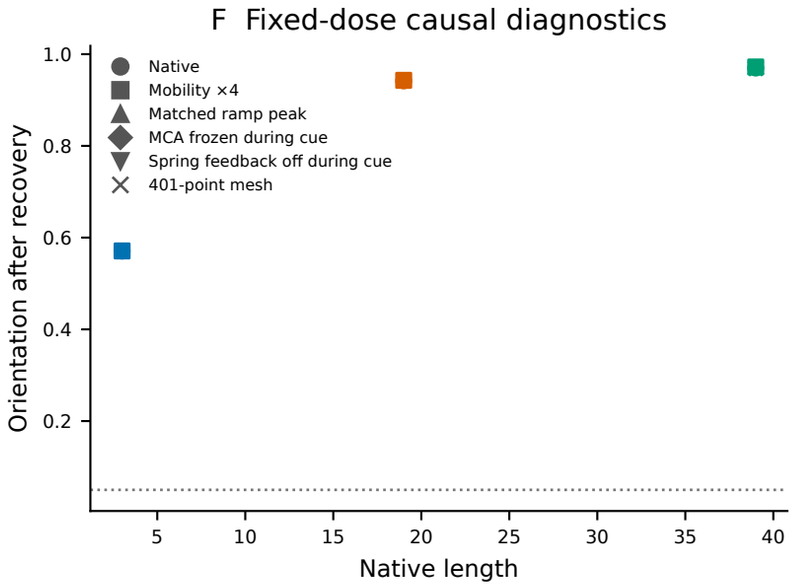






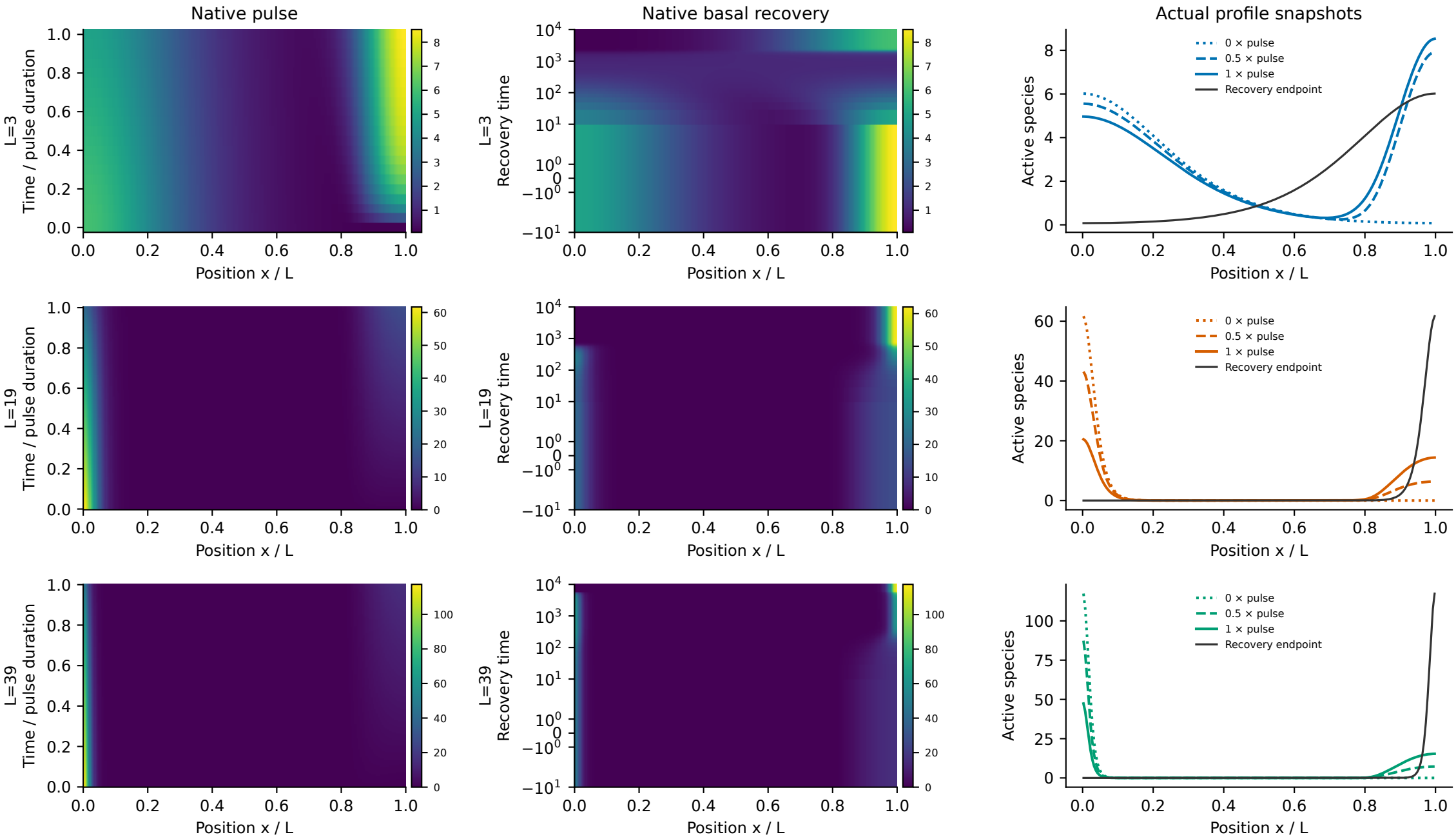






Amounts use excess above the current mid-cell median in each half; domain edges use that background plus half the initial pole contrast. Controls are fixed-dose diagnostics.

Goryachev · Relative width 20% · selected native spatial histories



Native scalar-selected pulses. Separate panel color scales preserve concentration changes; these are not independently verified steady states.

Goryachev · Constant physical slope

Mass sequestration in a concentrated self-amplifying pole

The incumbent pole stays about the same physical width but becomes much taller with length. Its inactive substrate at the opposite end falls sharply. The native added activation term is cue times inactive concentration, so equal patch peaks do not produce equal biochemical forcing. Slow competition and reconcentration of the conserved pool remain after cue initiation. The ramp peak increases with length because its physical slope is fixed. The growing ramp partly offsets the decline in initial inactive substrate, but removal/competition still slows. Matched-reference-peak ramps fail at the middle and longest native doses. Remeshing preserves the long-cell orientation classification at the selected orientation dose, while its numerical first moment changes.

Measured evidence

Old-width long/short=1.099; old-excess amount=27.05; opposite-end inactive ratio=0.0998; initial added activation-source ratio=1.297. Control failures at native doses: matched_gradient_peak 2/3, mesh_401 0/1, mobility_4x 0/3.

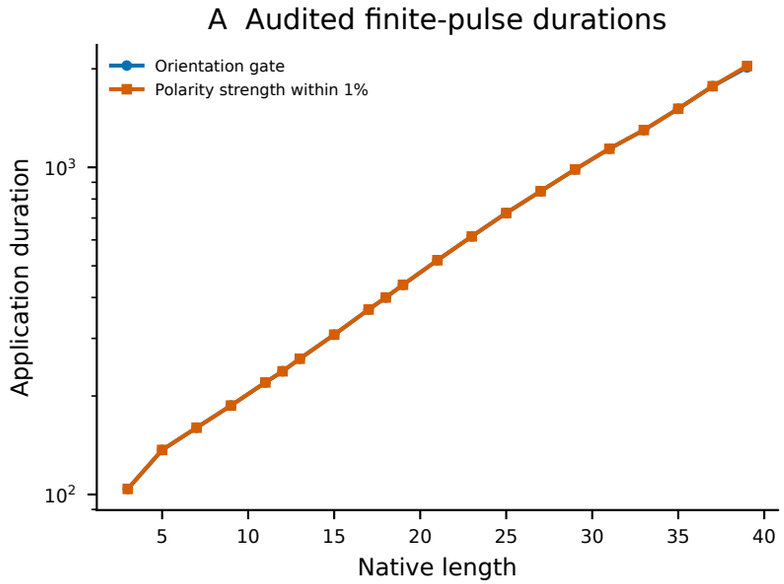
Testable prediction

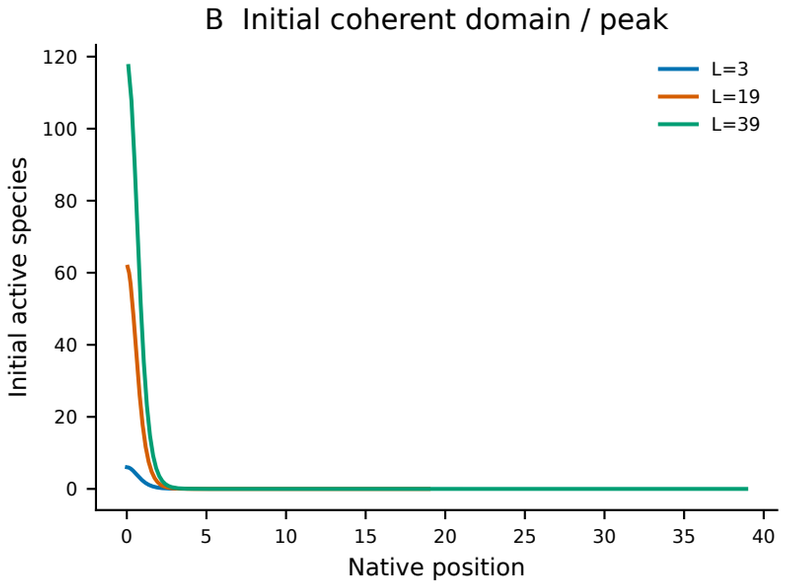
Measure old-pole height and opposite-end inactive abundance, rather than assuming a widening domain. Restoring substrate availability or using a separate fixed-total-material preparation should reduce the length dependence of initial patch activation; this proposed test is not part of the native comparison.

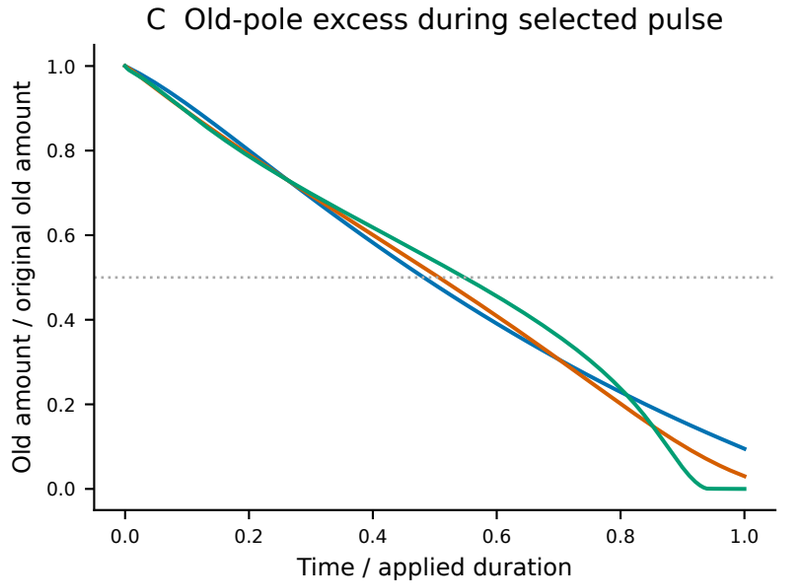
Scope of the inference

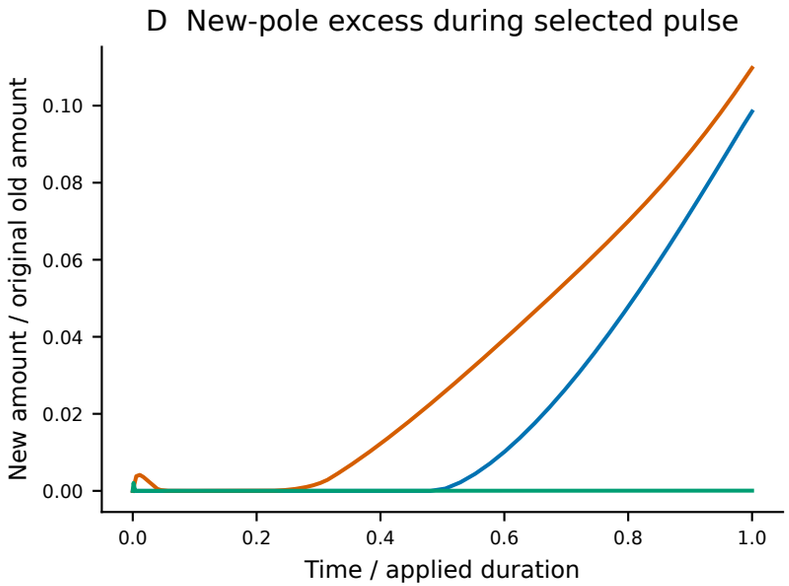
Trajectory and fixed-dose evidence in this native regime. Counterfactual thresholds were not fully searched; frozen-scale field flags and mesh sensitivity remain separate from scalar success.

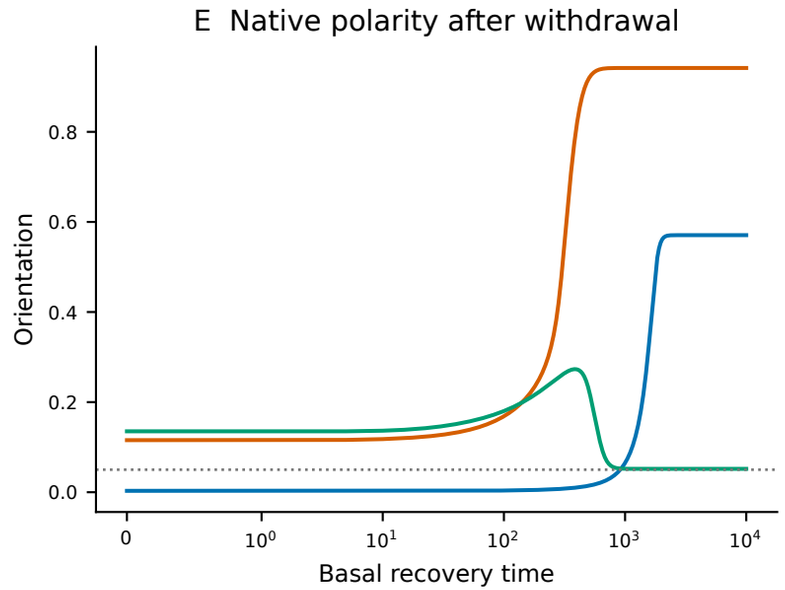
Goryachev · Constant physical slope

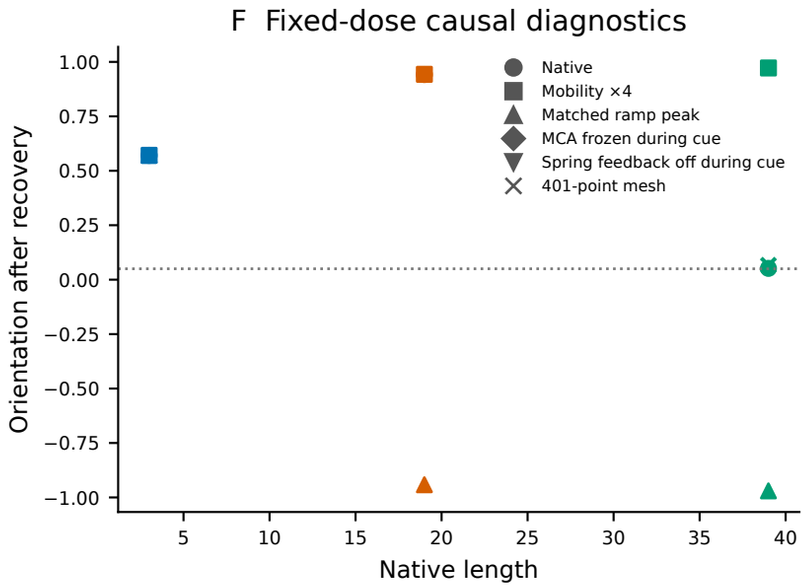




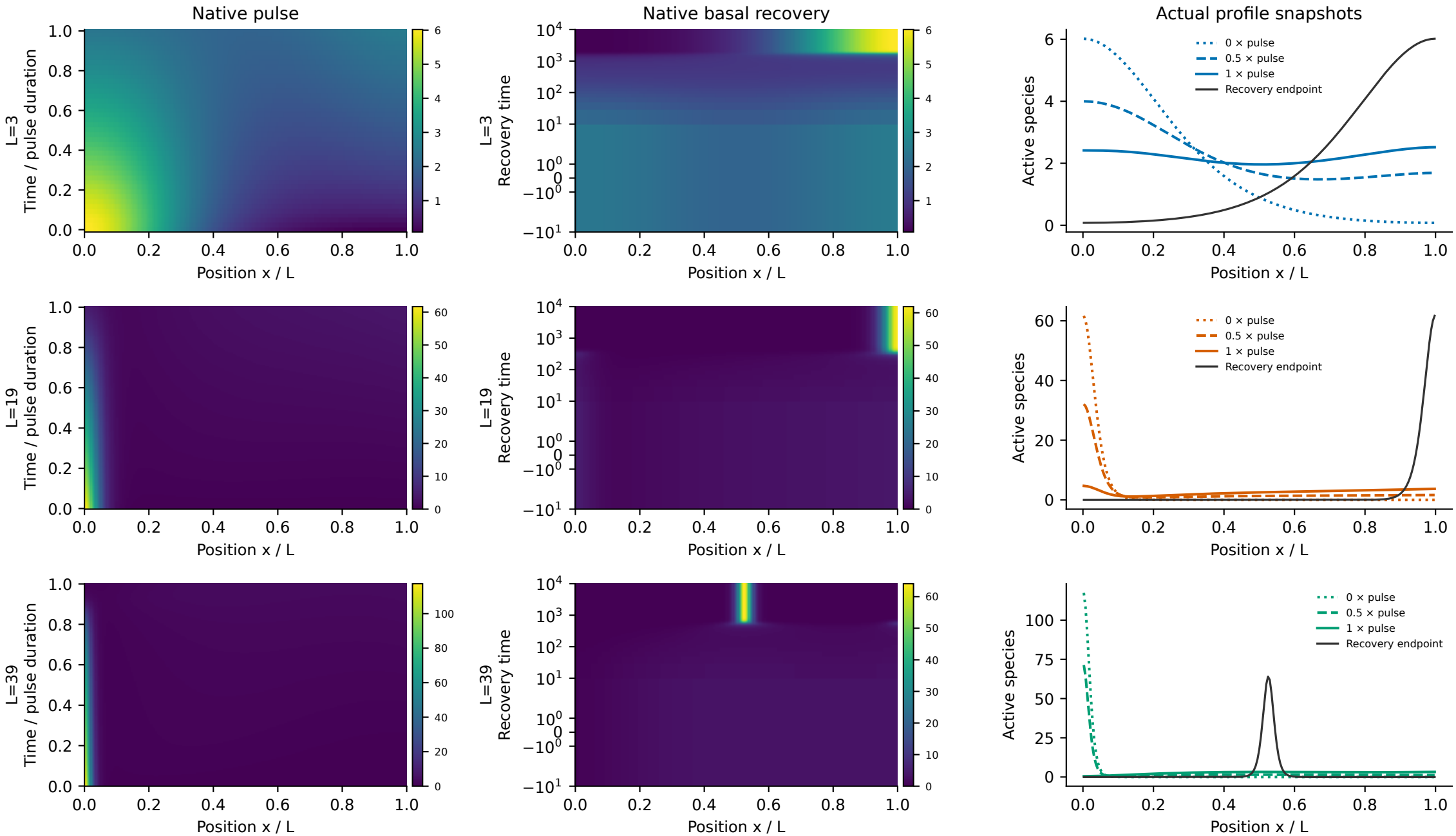








Amounts use excess above the current mid-cell median in each half; domain edges use that background plus half the initial pole contrast. Controls are fixed-dose diagnostics.



Native scalar-selected pulses. Separate panel color scales preserve concentration changes; these are not independently verified steady states.

deBelly · Fixed width 3

Rewriting a spatial adhesion/mechanical imprint

The fast Rac profile can respond while MCA still carries the old state. Freezing MCA redistribution during the pulse makes every sampled native-successful dose fail in this geometry when native recovery resumes. Changing unbound-MCA diffusion often removes retained polarization rather than simply speeding reversal, which also shows that diffusion helps determine memory strength. The fixed support covers a decreasing fraction of a longer cell. The longest native passing dose changes outcome on a 401-point mesh, so the exact longest-length threshold remains resolution sensitive.

Measured evidence

Old-width long/short=2.942; old-excess amount=3.896; opposite-end inactive ratio=not observed; initial added activation-source ratio=0.9954. Control failures at native doses: mca_frozen_during_pulse 4/4, mesh_401 1/1, mobility_4x 3/4.

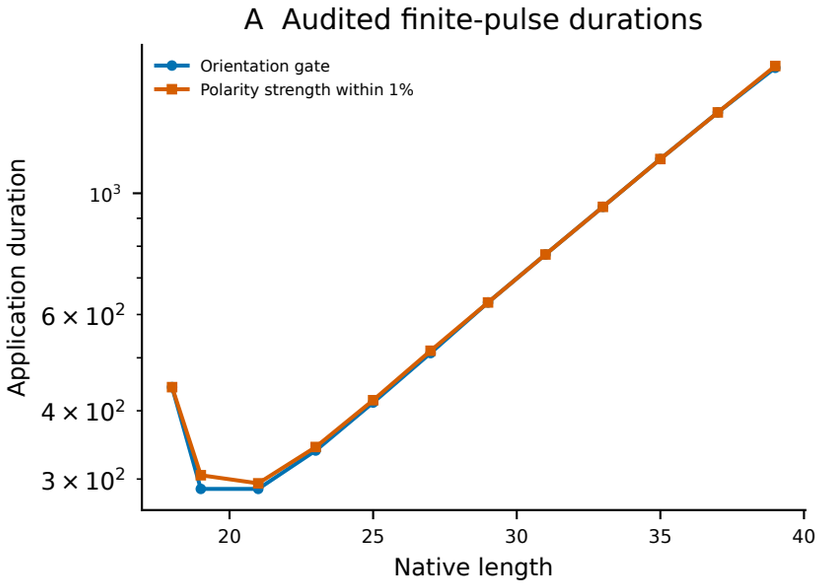
Testable prediction

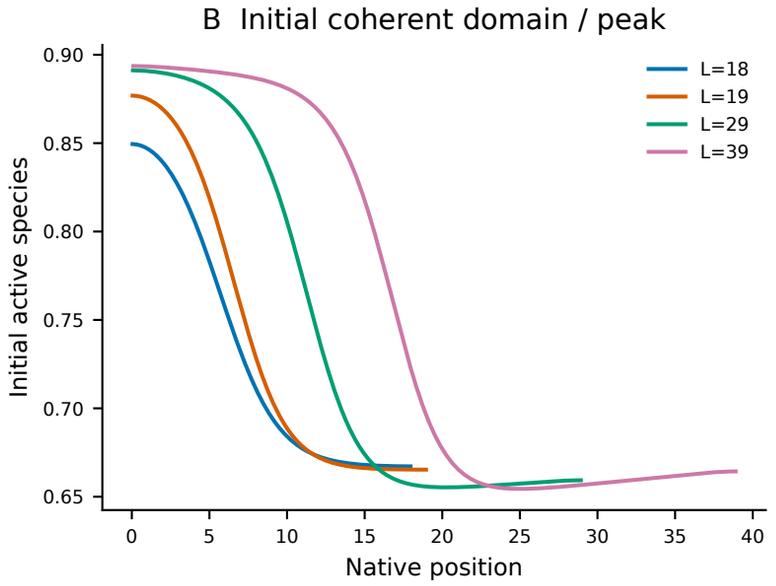
At matched Rac displacement, measure the bound/unbound MCA distribution and mechanical fields at cutoff. A Rac-only transient should relapse when the adhesion imprint is not rewritten; interventions on adhesion transport can change both the reversal threshold and the existence of persistent polarity.

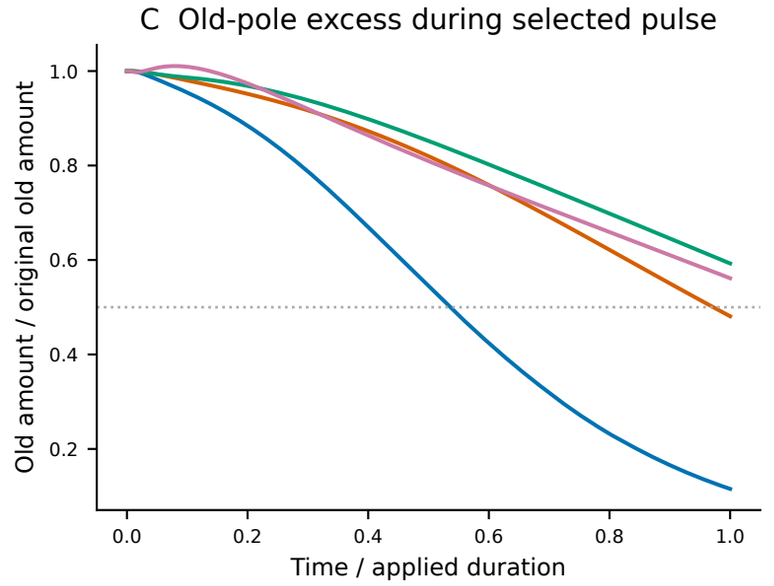
Scope of the inference

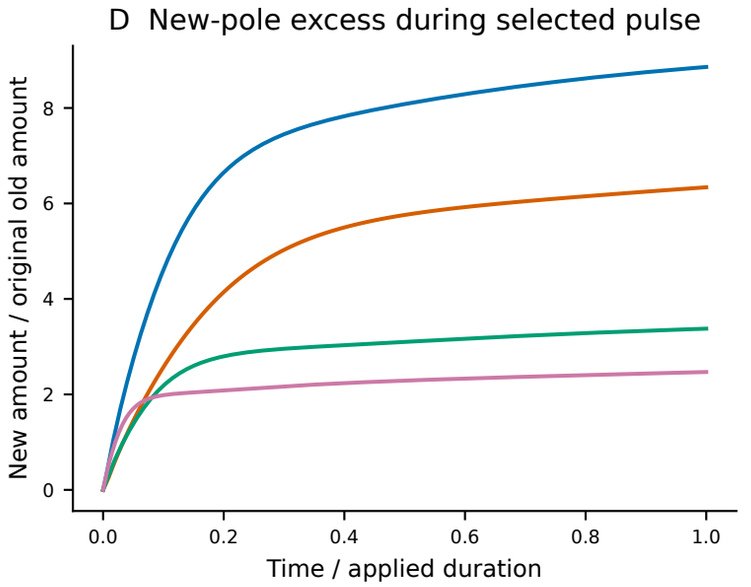
Trajectory and fixed-dose evidence in this native regime. Counterfactual thresholds were not fully searched; frozen-scale field flags and mesh sensitivity remain separate from scalar success.

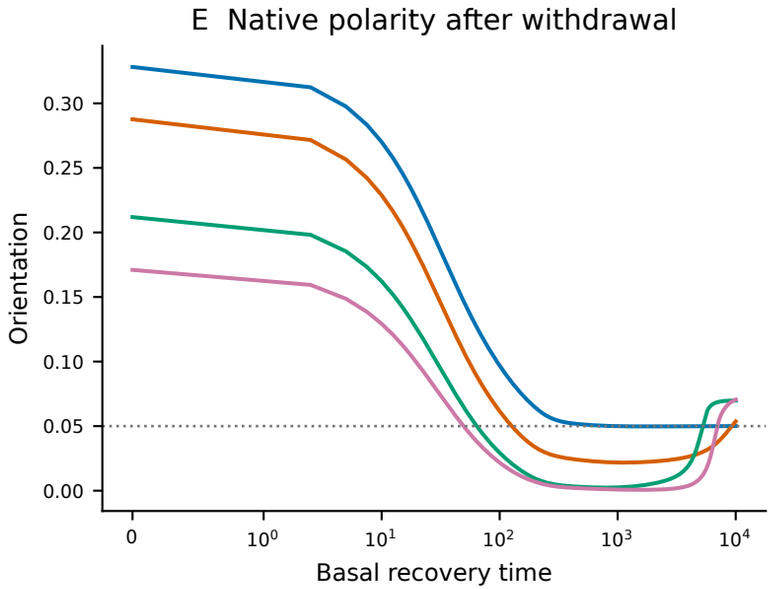
deBelly · Fixed width 3

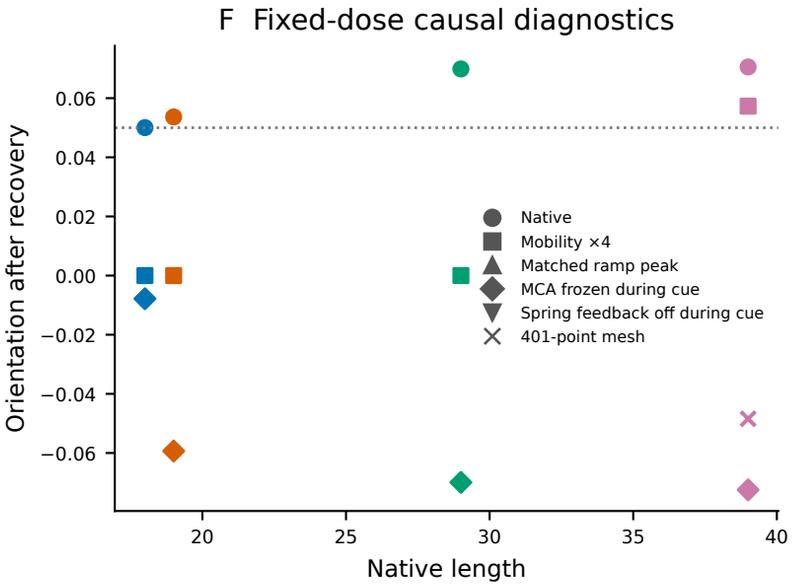






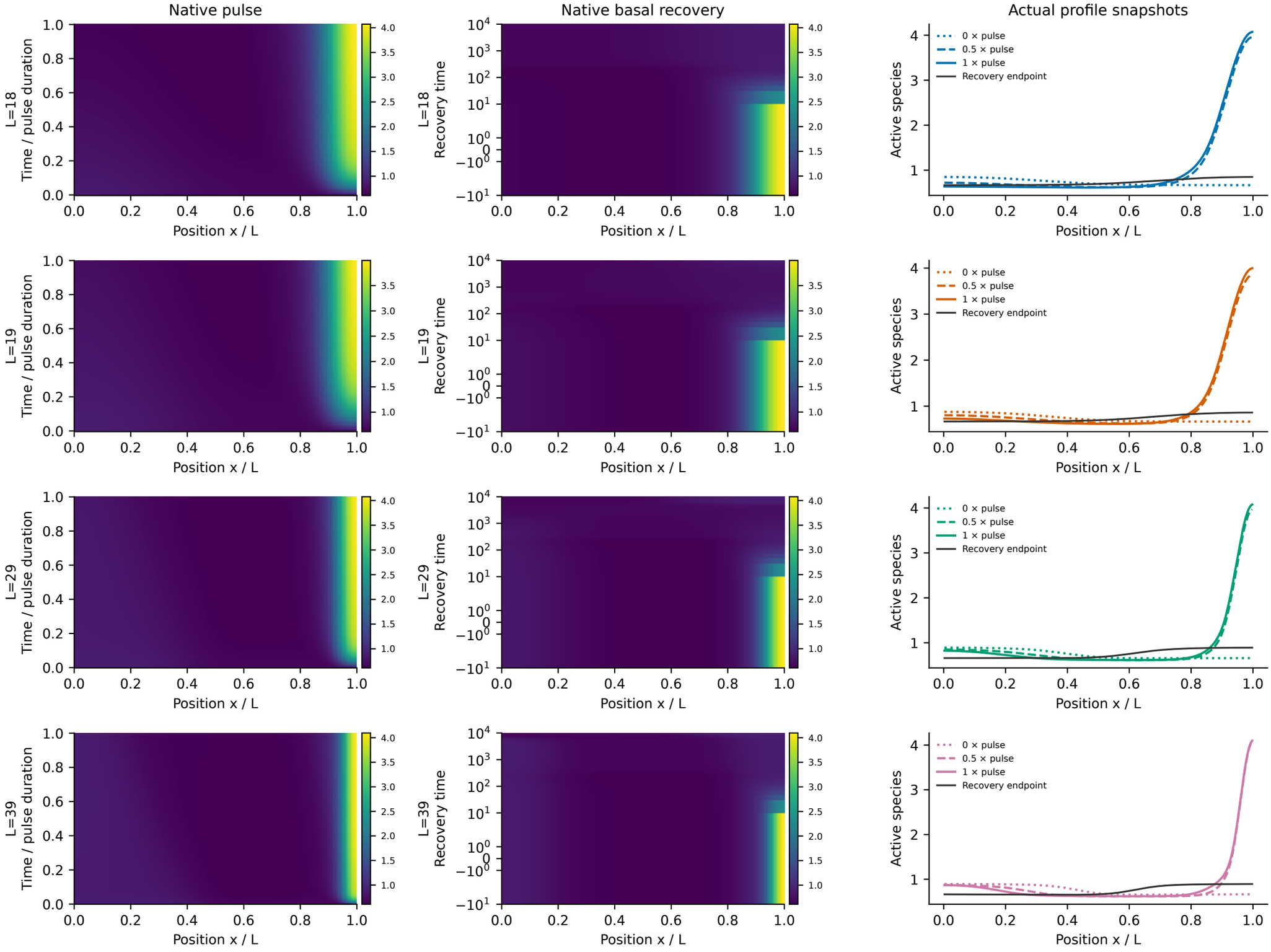




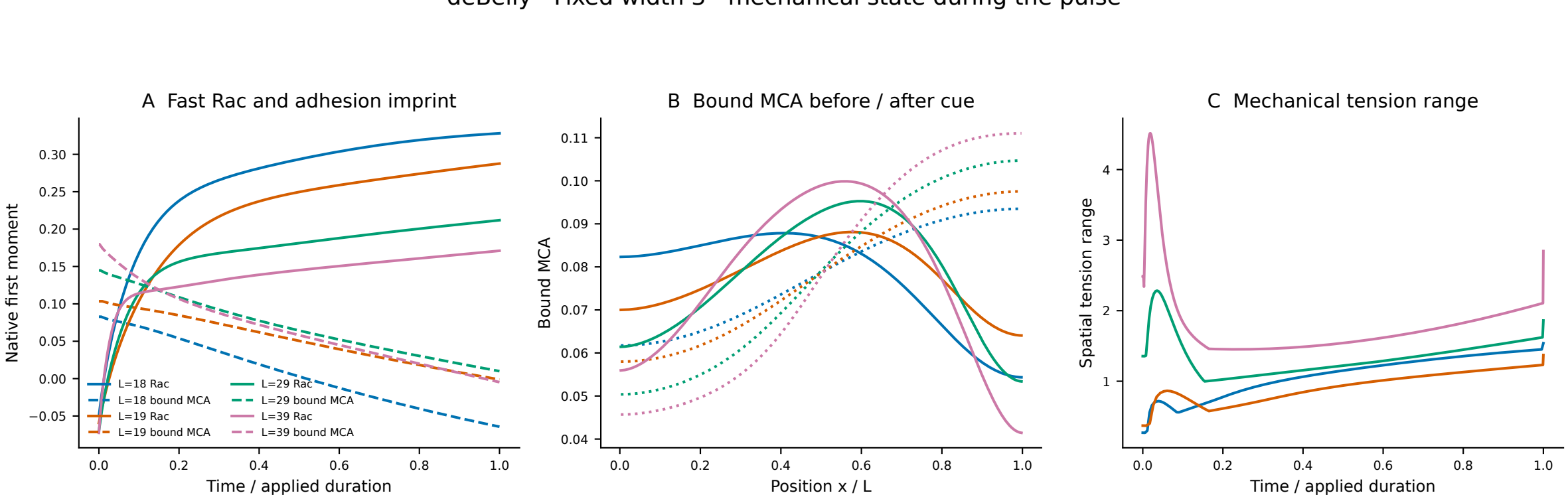


Amounts use excess above the current mid-cell median in each half; domain edges use that background plus half the initial pole contrast. Controls are fixed-dose diagnostics.

deBelly · Fixed width 3 · selected native spatial histories



Native scalar-selected pulses. Separate panel color scales preserve concentration changes; these are not independently verified steady states.


deBelly · Fixed width 3 · mechanical state during the pulse
A Fast Rac and adhesion imprint
Native first moment
Time / applied duration
L=18 Rac
L=18 bound MCA
L=19 Rac
L=19 bound MCA
L=29 Rac
L=29 bound MCA
L=39 Rac
L=39 bound MCA
B Bound MCA before / after cue
Bound MCA
Position x / L
C Mechanical tension range
Spatial tension range
Time / applied duration
Solid: Rac or cutoff profile; dashed: bound-MCA moment; dotted: initial MCA profile. MCA and Rac can have opposite spatial moments. Frozen-MCA controls preserve the original adhesion fields during cue only.

deBelly · Relative width 20%

Rewriting a spatial adhesion/mechanical imprint

The fast Rac profile can respond while MCA still carries the old state. Freezing MCA redistribution during the pulse makes every sampled native-successful dose fail in this geometry when native recovery resumes. Changing unbound-MCA diffusion often removes retained polarization rather than simply speeding reversal, which also shows that diffusion helps determine memory strength. The cue keeps the same footprint fraction and peak as length grows. Cue-only MCA freezing fails at every sampled length, and faster unbound-MCA diffusion removes polarization at the shorter doses.

Measured evidence

Old-width long/short=2.942; old-excess amount=3.896; opposite-end inactive ratio=not observed; initial added activation-source ratio=0.9974. Control failures at native doses: mca_frozen_during_pulse 4/4, mesh_401 0/1, mobility_4x 3/4.

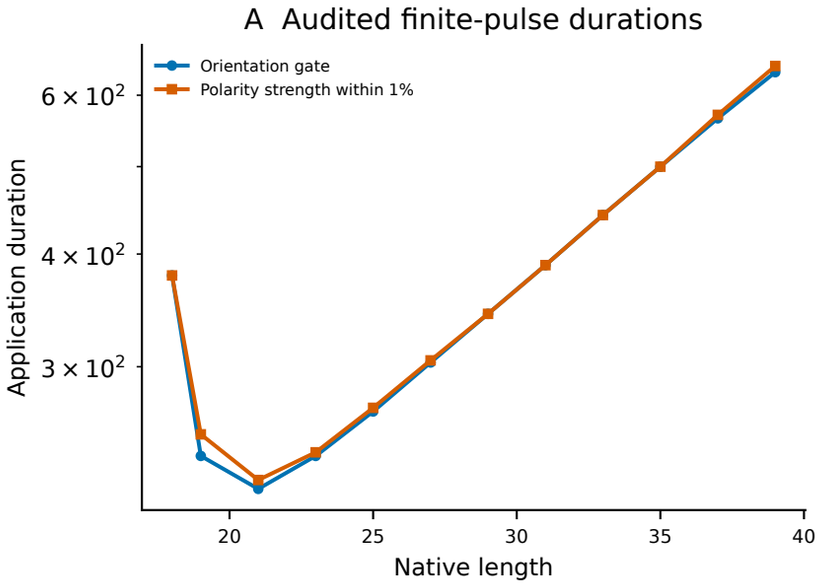
Testable prediction

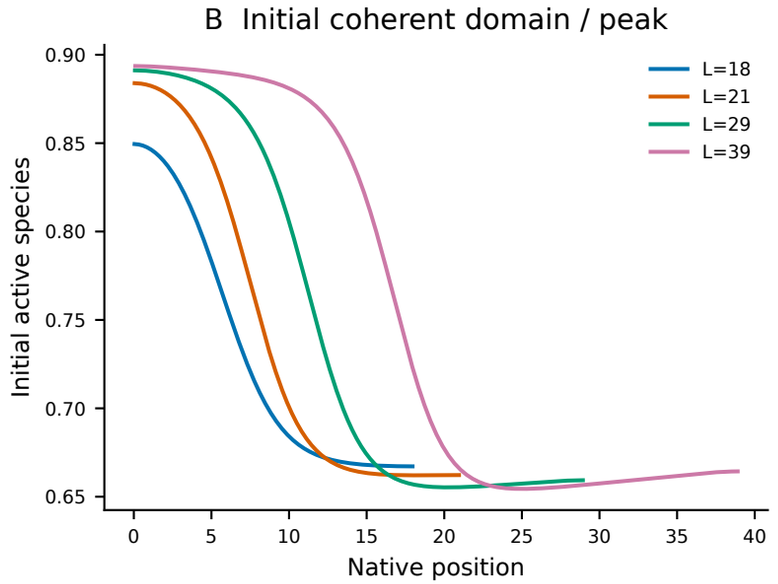
At matched Rac displacement, measure the bound/unbound MCA distribution and mechanical fields at cutoff. A Rac-only transient should relapse when the adhesion imprint is not rewritten; interventions on adhesion transport can change both the reversal threshold and the existence of persistent polarity.

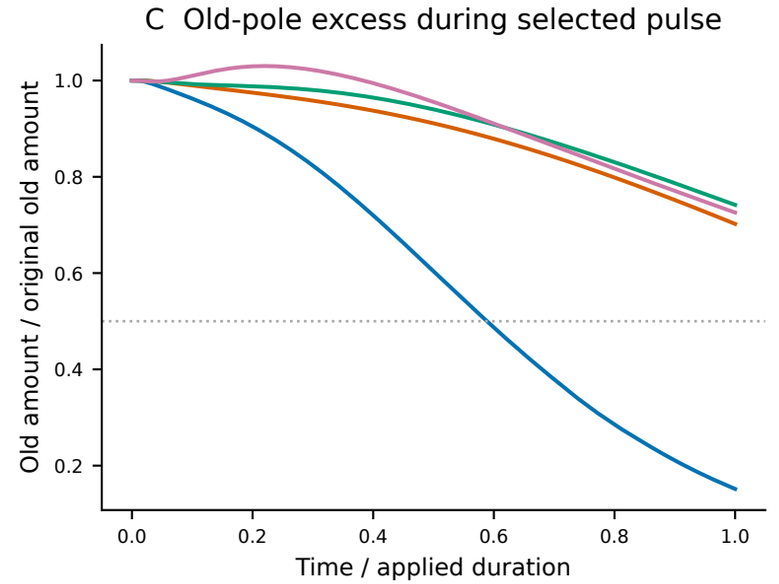
Scope of the inference

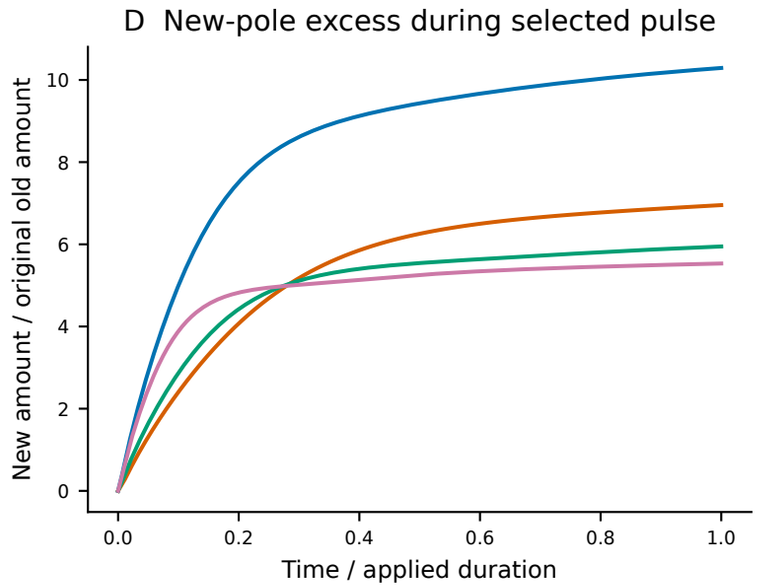
Trajectory and fixed-dose evidence in this native regime. Counterfactual thresholds were not fully searched; frozen-scale field flags and mesh sensitivity remain separate from scalar success.

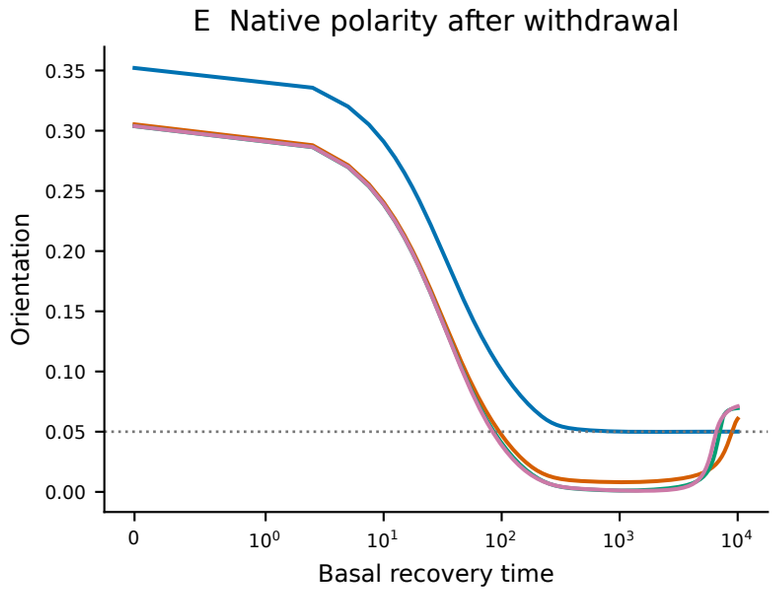
deBelly · Relative width 20%

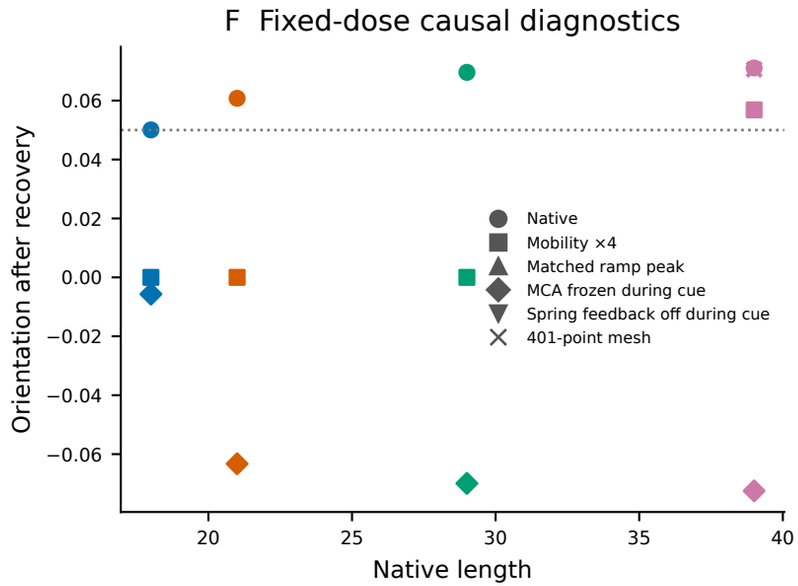




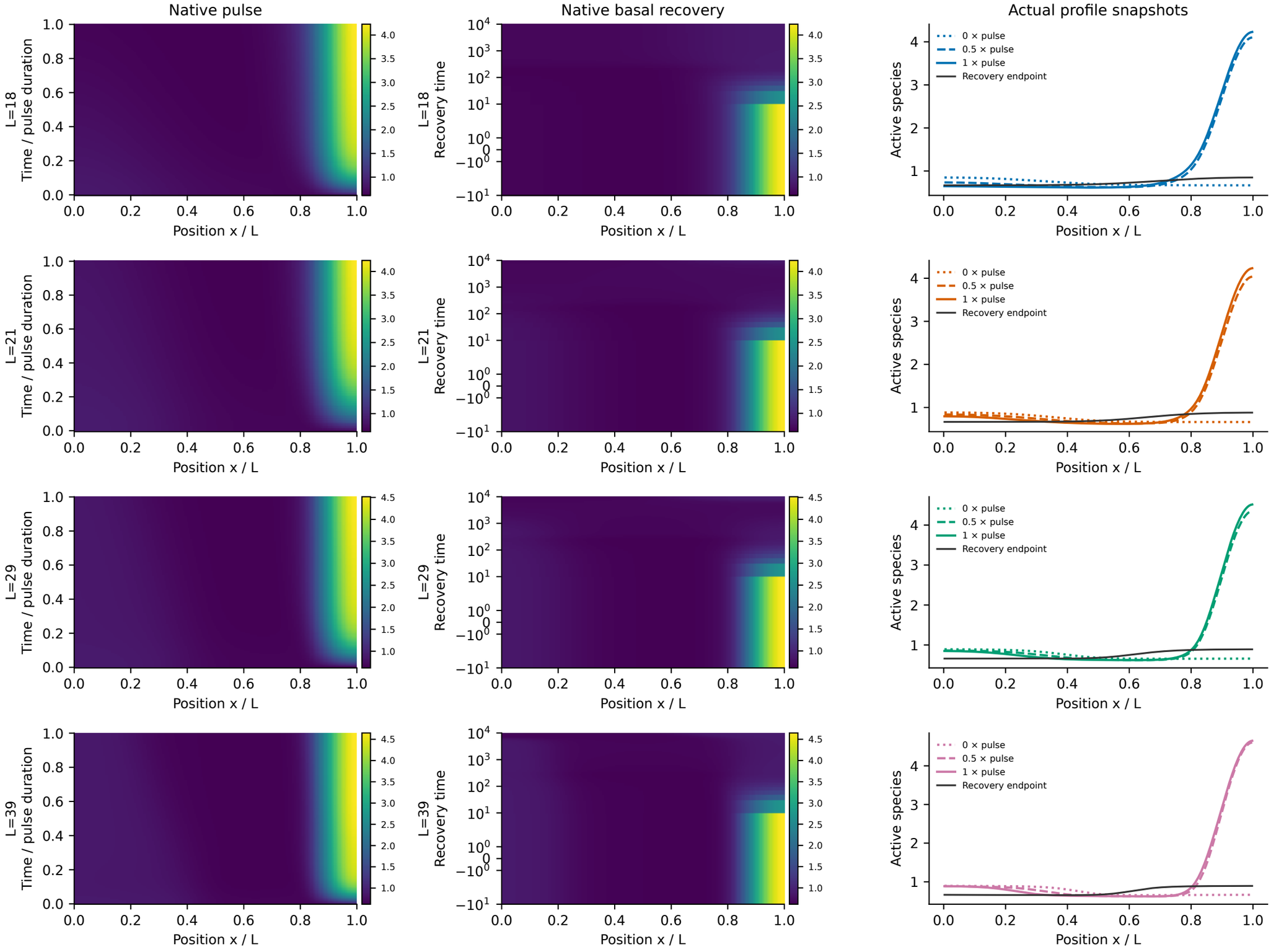




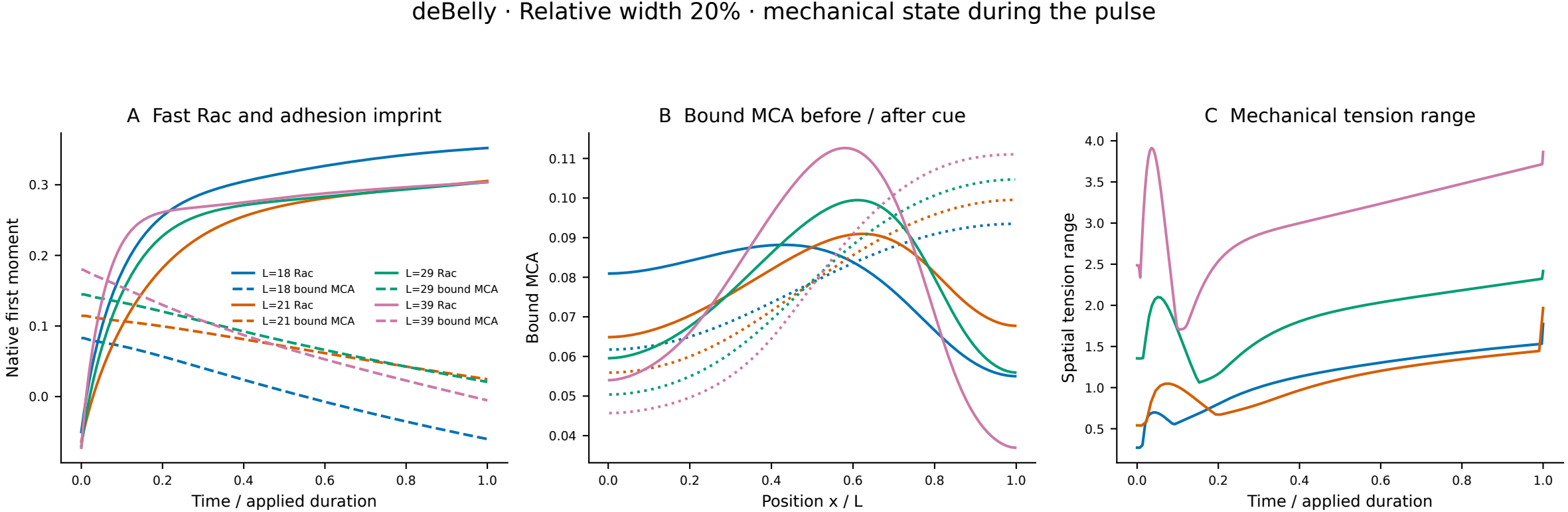




Amounts use excess above the current mid-cell median in each half; domain edges use that background plus half the initial pole contrast. Controls are fixed-dose diagnostics.



Native scalar-selected pulses. Separate panel color scales preserve concentration changes; these are not independently verified steady states.



Solid: Rac or cutoff profile; dashed: bound-MCA moment; dotted: initial MCA profile. MCA and Rac can have opposite spatial moments. Frozen-MCA controls preserve the original adhesion fields during cue only.

deBelly · Constant physical slope

Rewriting a spatial adhesion/mechanical imprint

The fast Rac profile can respond while MCA still carries the old state. Freezing MCA redistribution during the pulse makes every sampled native-successful dose fail in this geometry when native recovery resumes. Changing unbound-MCA diffusion often removes retained polarization rather than simply speeding reversal, which also shows that diffusion helps determine memory strength. The ramp peak increases with length because its physical slope is fixed. All sampled matched-reference-peak doses still pass orientation, so this control does not identify a critical ramp-peak bottleneck here. MCA freezing fails throughout. The short arm is also conditioned by the scalar gate and relaxation near the preparation-eligibility boundary; native field flags remain explicit.

Measured evidence

Old-width long/short=2.942; old-excess amount=3.896; opposite-end inactive ratio=not observed; initial added activation-source ratio=2.161. Control failures at native doses: matched_gradient_peak 0/4, mca_frozen_during_pulse 4/4, mesh_401 0/1, mobility_4x 4/4.

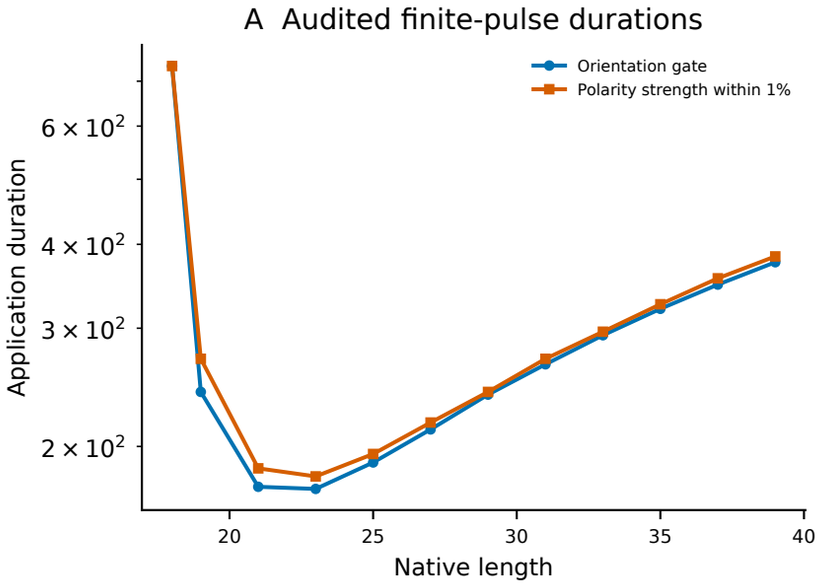
Testable prediction

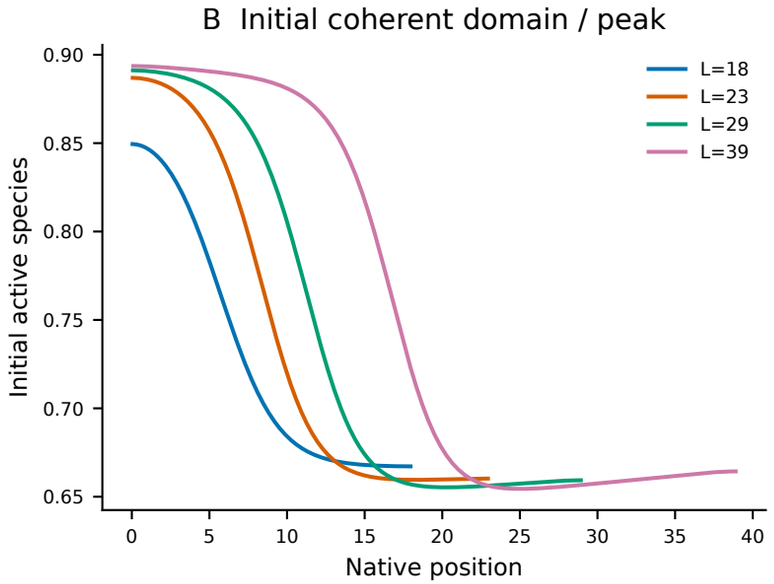
At matched Rac displacement, measure the bound/unbound MCA distribution and mechanical fields at cutoff. A Rac-only transient should relapse when the adhesion imprint is not rewritten; interventions on adhesion transport can change both the reversal threshold and the existence of persistent polarity.

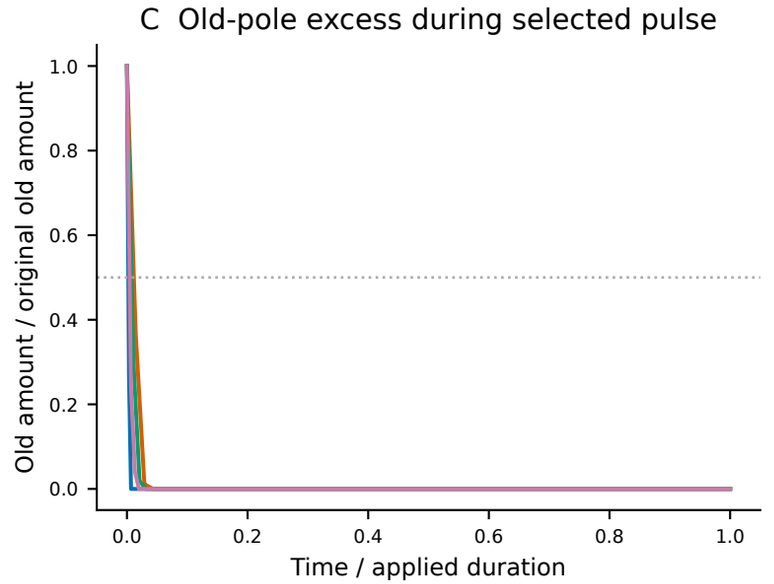
Scope of the inference

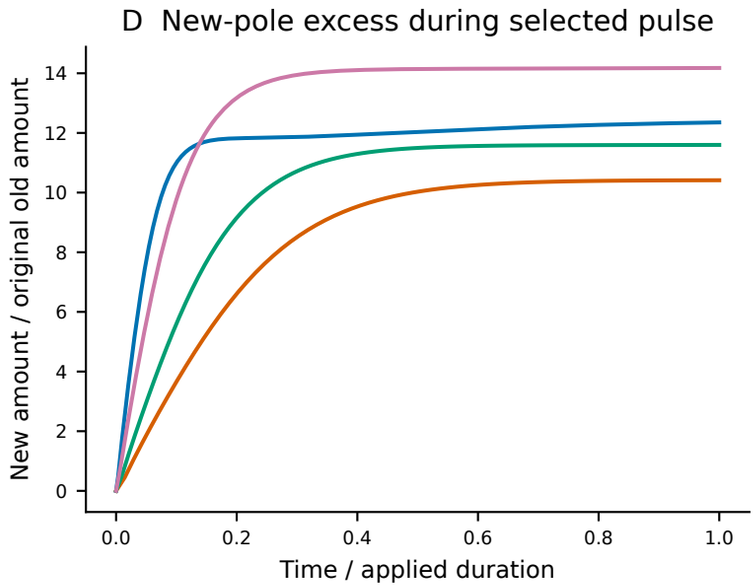
Trajectory and fixed-dose evidence in this native regime. Counterfactual thresholds were not fully searched; frozen-scale field flags and mesh sensitivity remain separate from scalar success.

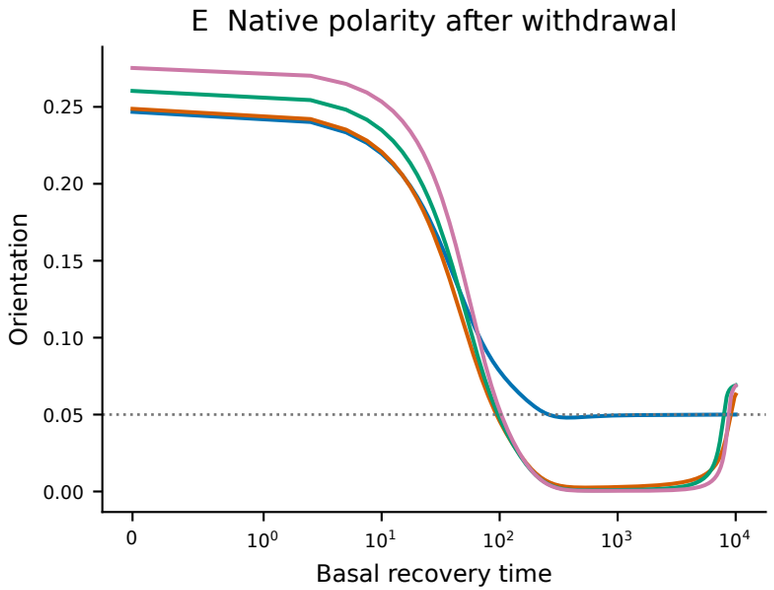
deBelly · Constant physical slope

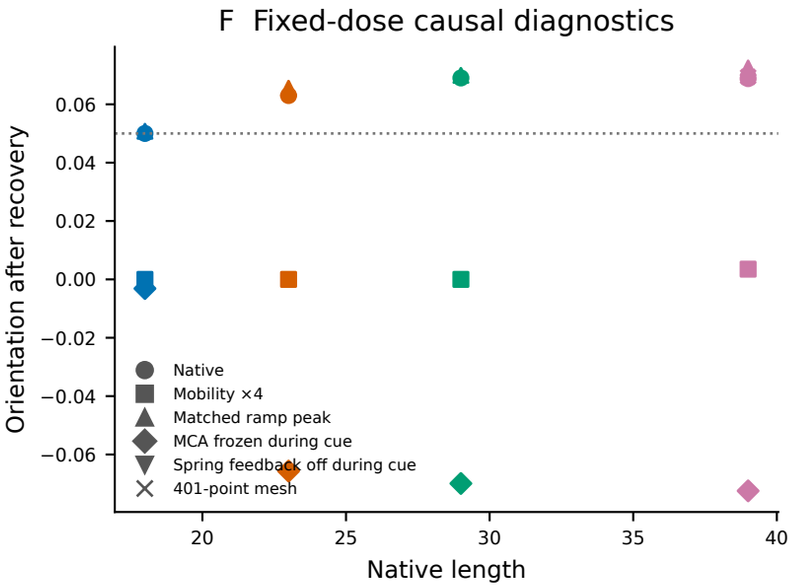




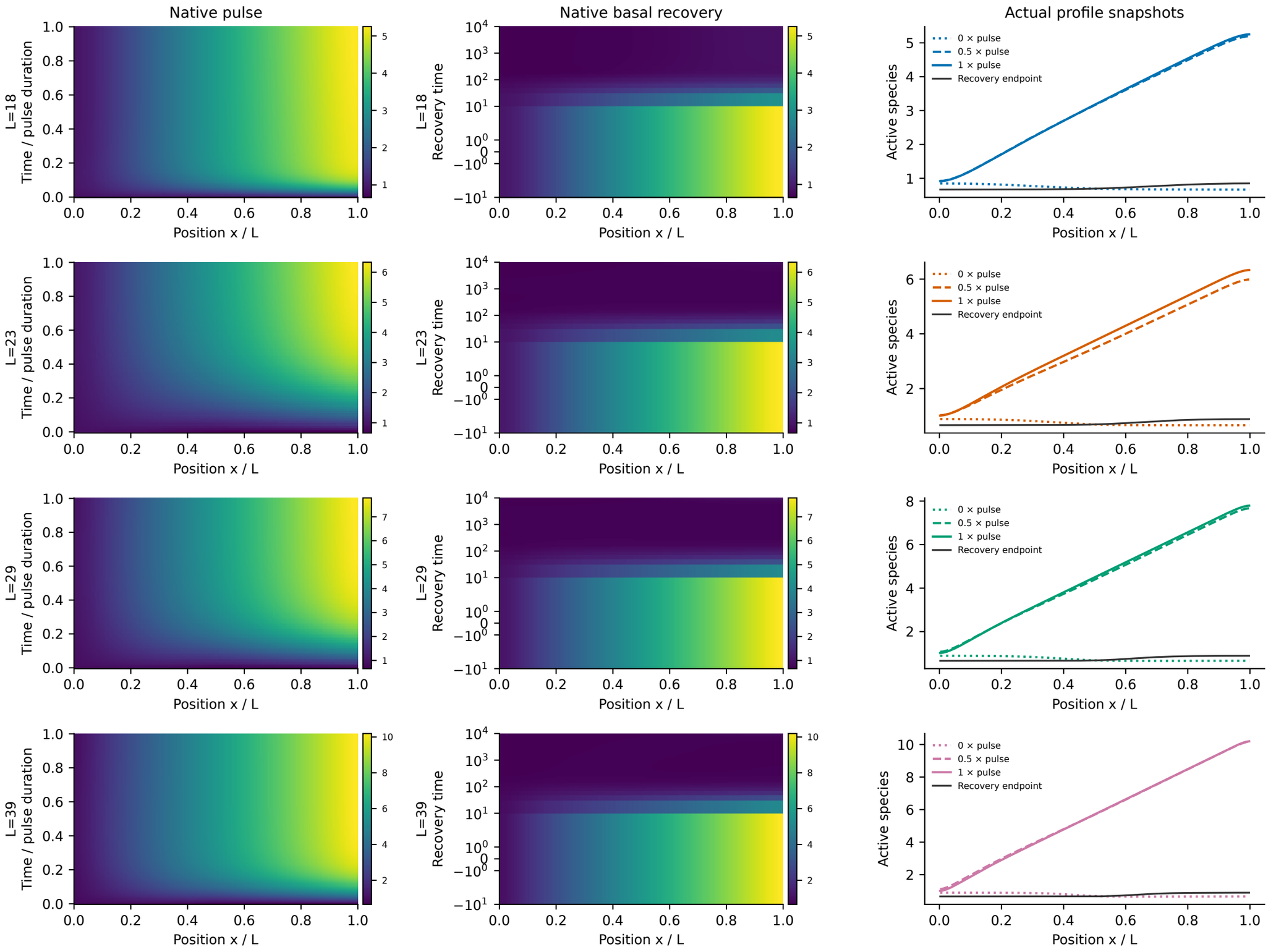


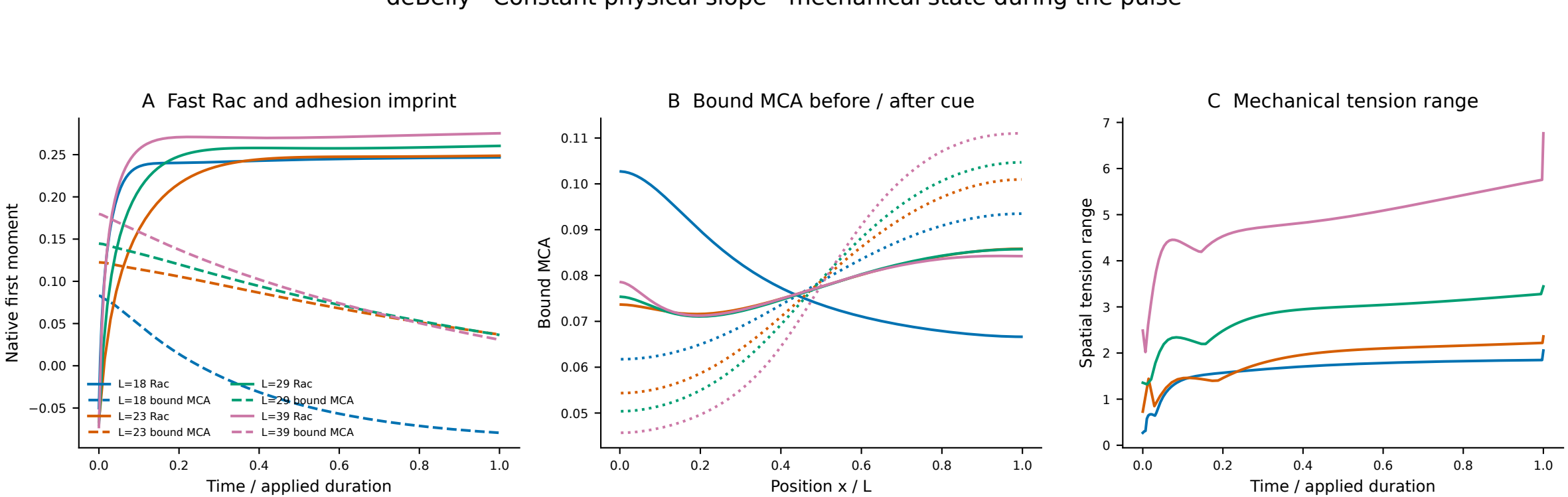






Amounts use excess above the current mid-cell median in each half; domain edges use that background plus half the initial pole contrast. Controls are fixed-dose diagnostics.


deBelly · Constant physical slope · selected native spatial histories
Native pulse
Native basal recovery
Actual profile snapshots
L=18
L=23
L=29
L=39
Time / pulse duration
Recovery time
Active species
Position x / L
0 × pulse
0.5 × pulse
1 × pulse
Recovery endpoint
Native scalar-selected pulses. Separate panel color scales preserve concentration changes; these are not independently verified steady states.



Solid: Rac or cutoff profile; dashed: bound-MCA moment; dotted: initial MCA profile. MCA and Rac can have opposite spatial moments. Frozen-MCA controls preserve the original adhesion fields during cue only.

Holmes 3 · Fixed width 3

Spatial coverage plus coupled Rac/Rho domain competition

Initial Rac plateau height and inactive availability are almost length independent, but the Rac domain expands. Mutual inhibition with Rho and the conserved pools supports competing spatial domains. A localized right seed need not overwrite the incumbent, even when its right activity rises quickly. The fixed support covers a decreasing fraction of a longer cell. At L=28 and 50, fourfold inactive mobility does not rescue the maximum native tested pulse; relative support is more effective than this mobility change.

Measured evidence

Old-width long/short=2.587; old-excess amount=2.989; opposite-end inactive ratio=0.9996; initial added activation-source ratio=0.9987. Control failures at native doses: mesh_401 1/1, mobility_4x 2/4.

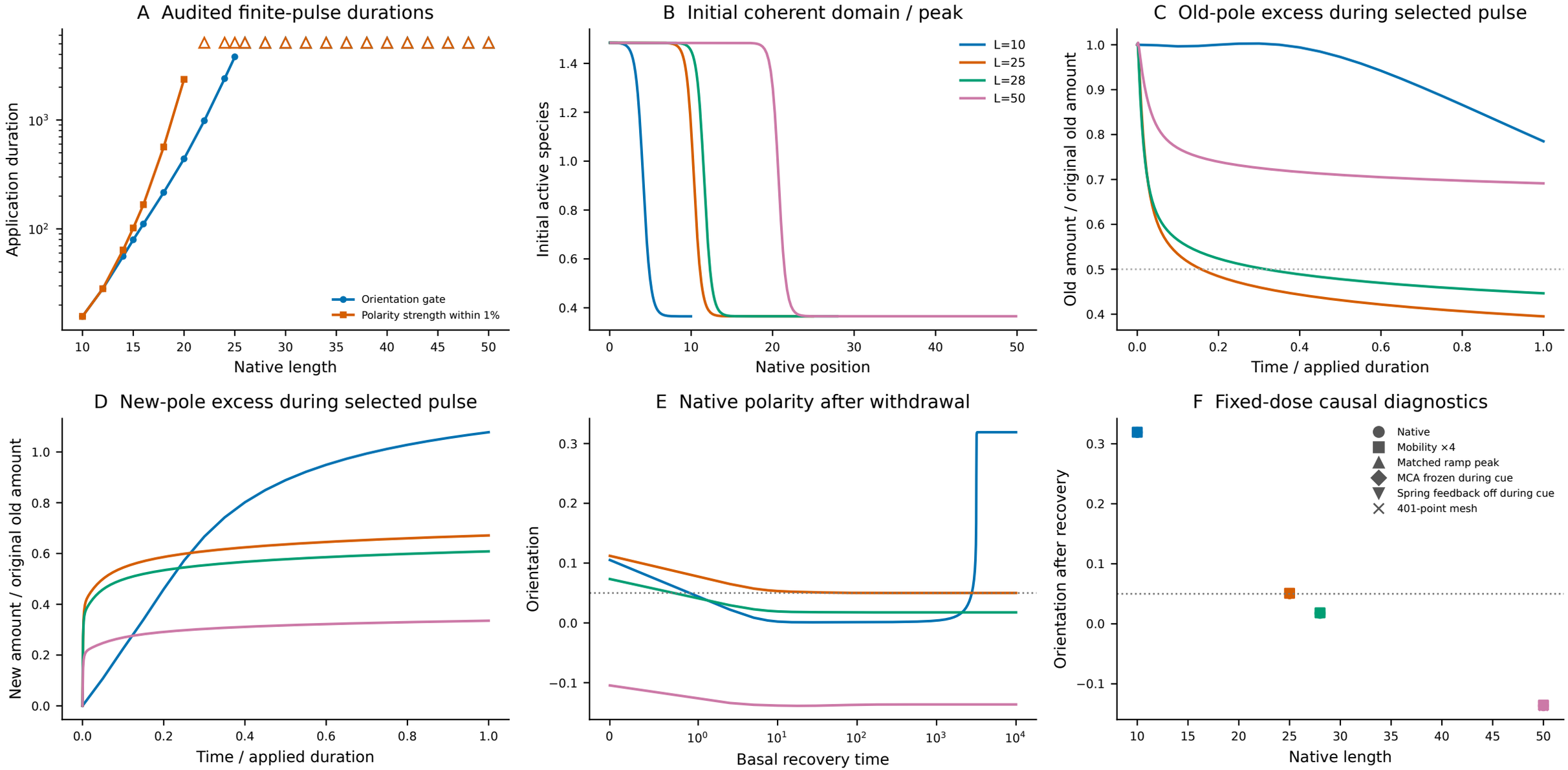
Testable prediction

Track both Rac and Rho boundaries and inactive-pool gradients. Broader support at the same peak should rescue spatial-coverage failures; faster inactive transport can improve gradient recovery without removing every localized-patch failure.

Scope of the inference

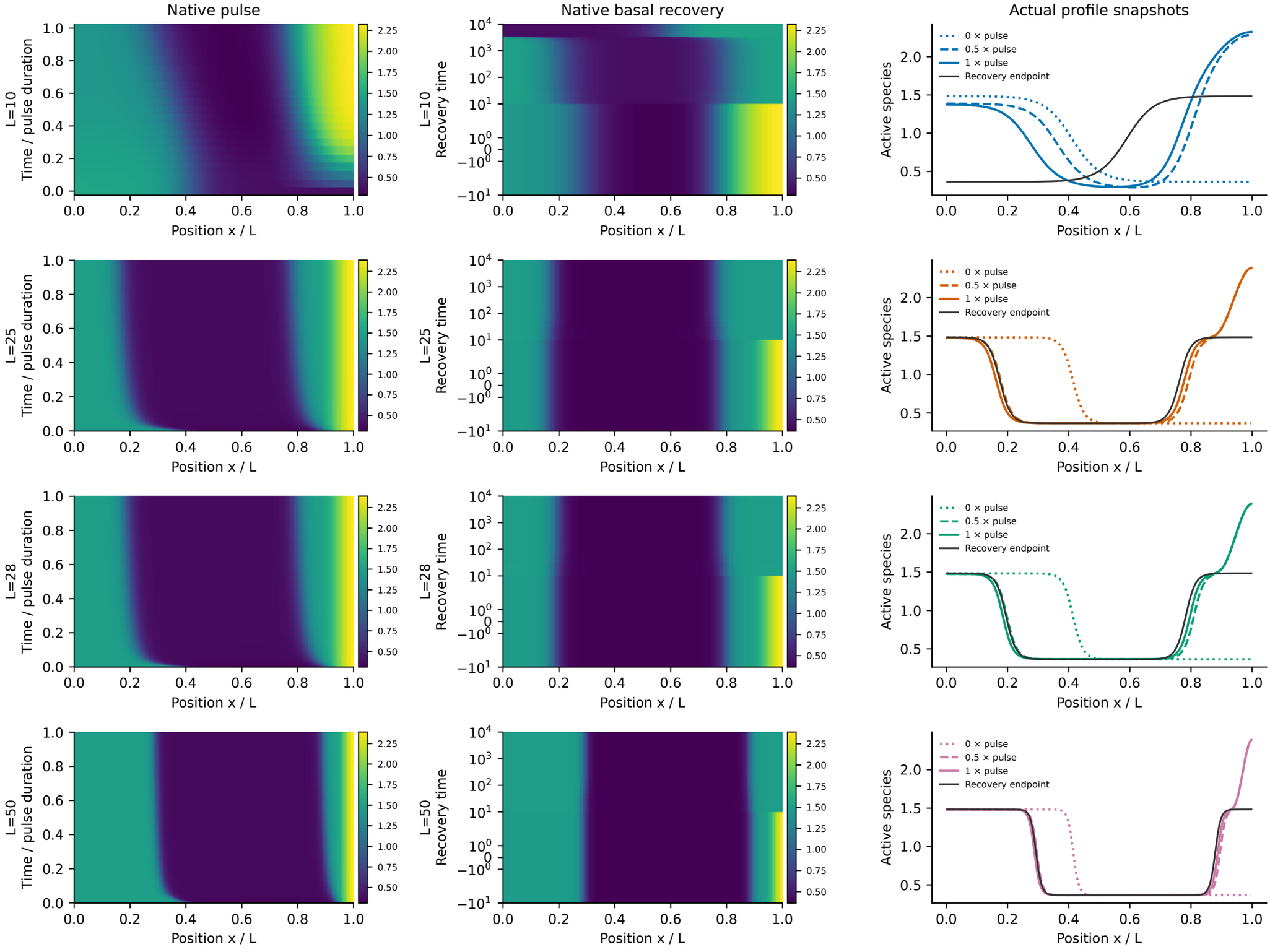
Trajectory and fixed-dose evidence in this native regime. Counterfactual thresholds were not fully searched; frozen-scale field flags and mesh sensitivity remain separate from scalar success.

Holmes 3 · Fixed width 3



Amounts use excess above the current mid-cell median in each half; domain edges use that background plus half the initial pole contrast. Controls are fixed-dose diagnostics.

Holmes 3 · Fixed width 3 · selected native spatial histories



Native scalar-selected pulses. Separate panel color scales preserve concentration changes; these are not independently verified steady states.

Holmes 3 · Relative width 20%

Spatial coverage plus coupled Rac/Rho domain competition

Initial Rac plateau height and inactive availability are almost length independent, but the Rac domain expands. Mutual inhibition with Rho and the conserved pools supports competing spatial domains. A localized right seed need not overwrite the incumbent, even when its right activity rises quickly. The cue keeps the same footprint fraction and peak as length grows. Orientation passes at all lengths, while many long cells remain outside the original-strength gate. This is a domain/shape and readout distinction, not absence of rightward response.

Measured evidence

Old-width long/short=5.19; old-excess amount=6.067; opposite-end inactive ratio=0.9995; initial added activation-source ratio=0.9995. Control failures at native doses: mesh_401 0/1, mobility_4x 1/3.

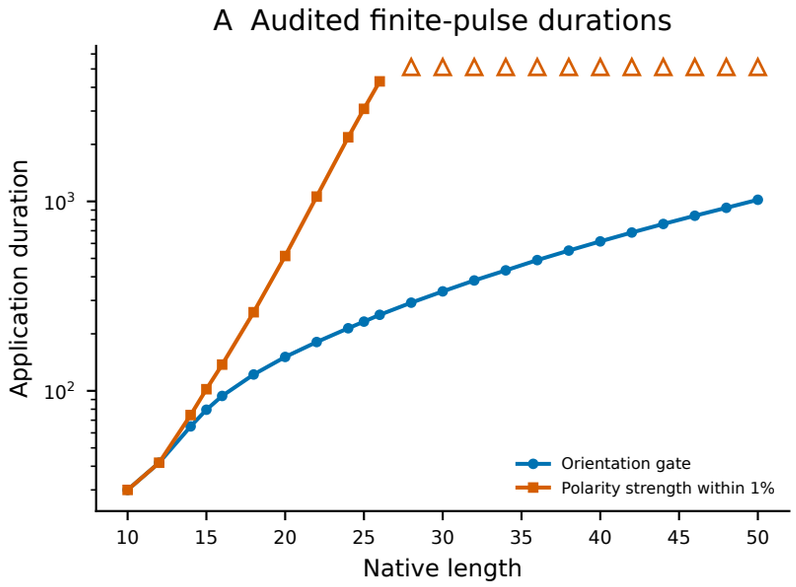
Testable prediction

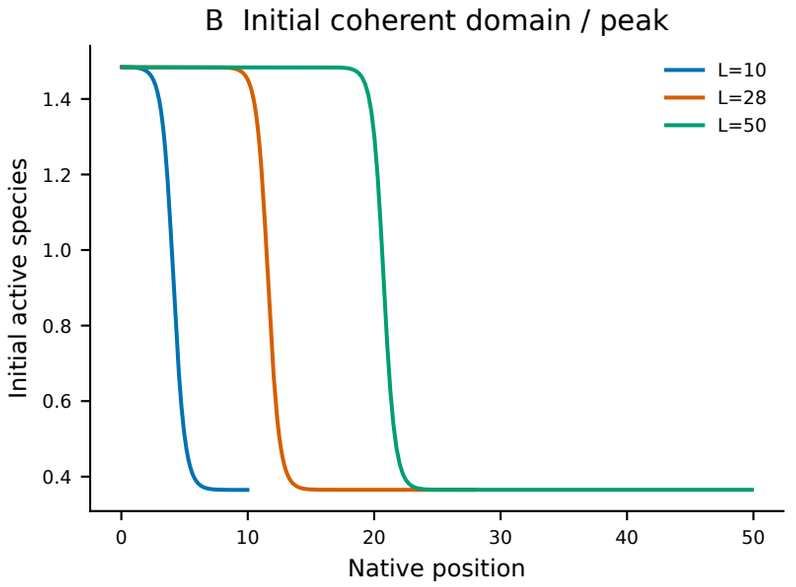
Track both Rac and Rho boundaries and inactive-pool gradients. Broader support at the same peak should rescue spatial-coverage failures; faster inactive transport can improve gradient recovery without removing every localized-patch failure.

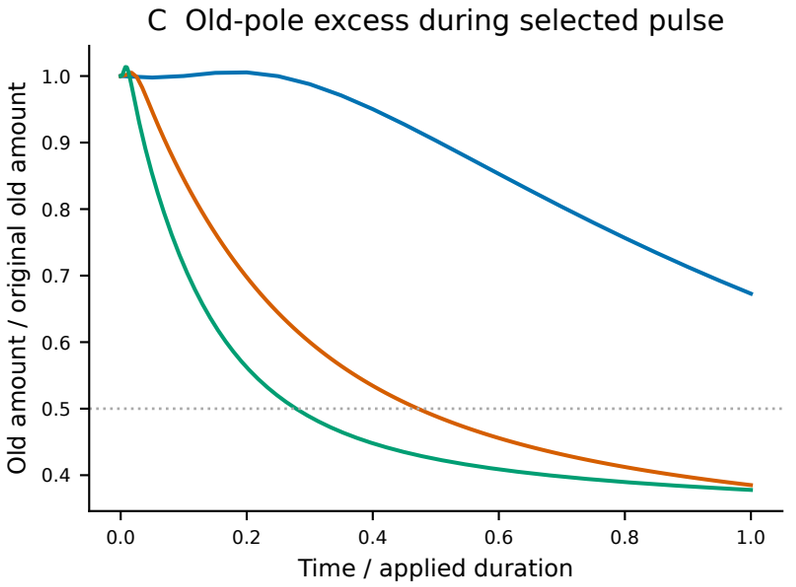
Scope of the inference

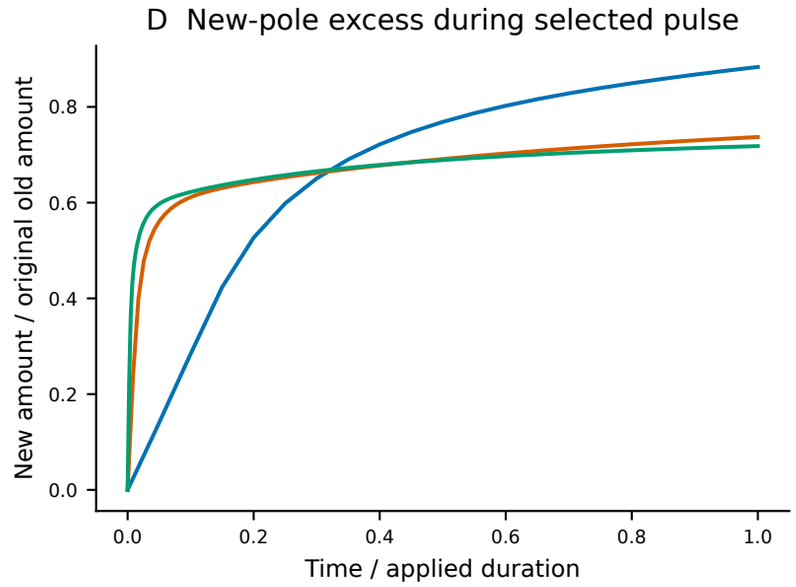
Trajectory and fixed-dose evidence in this native regime. Counterfactual thresholds were not fully searched; frozen-scale field flags and mesh sensitivity remain separate from scalar success.

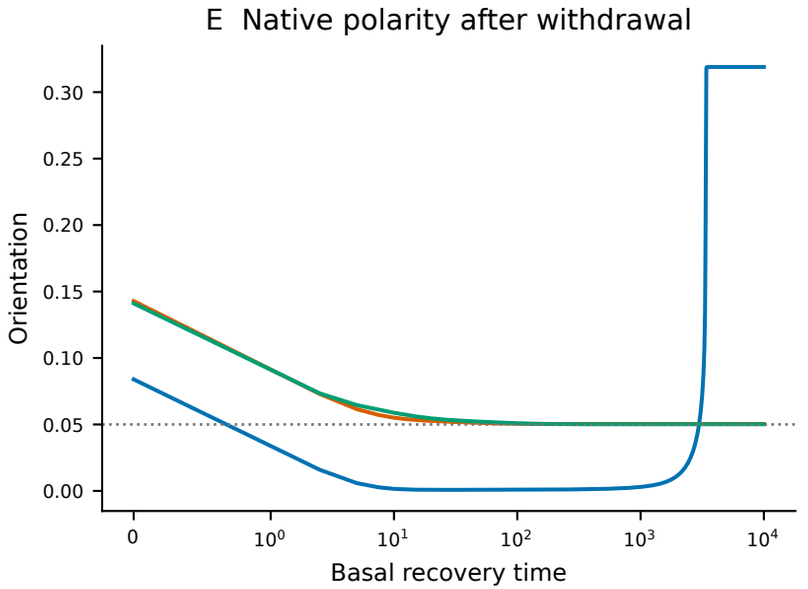
Holmes 3 · Relative width 20%

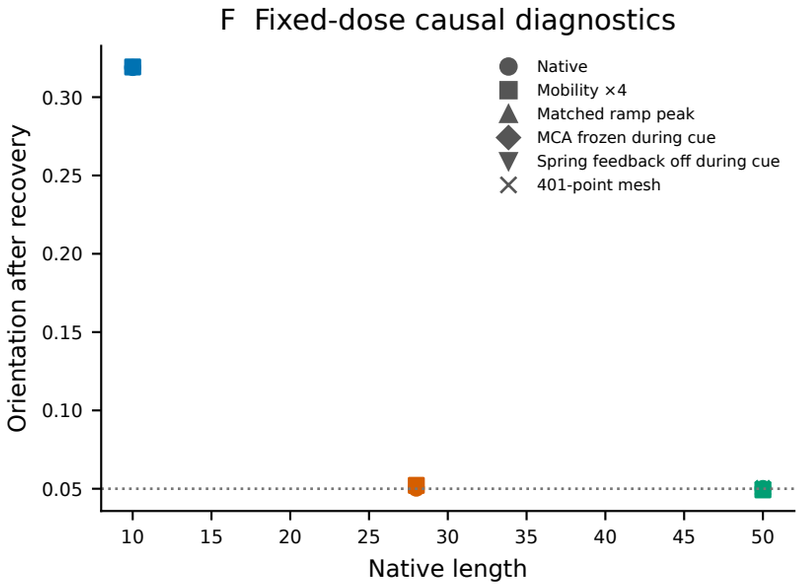




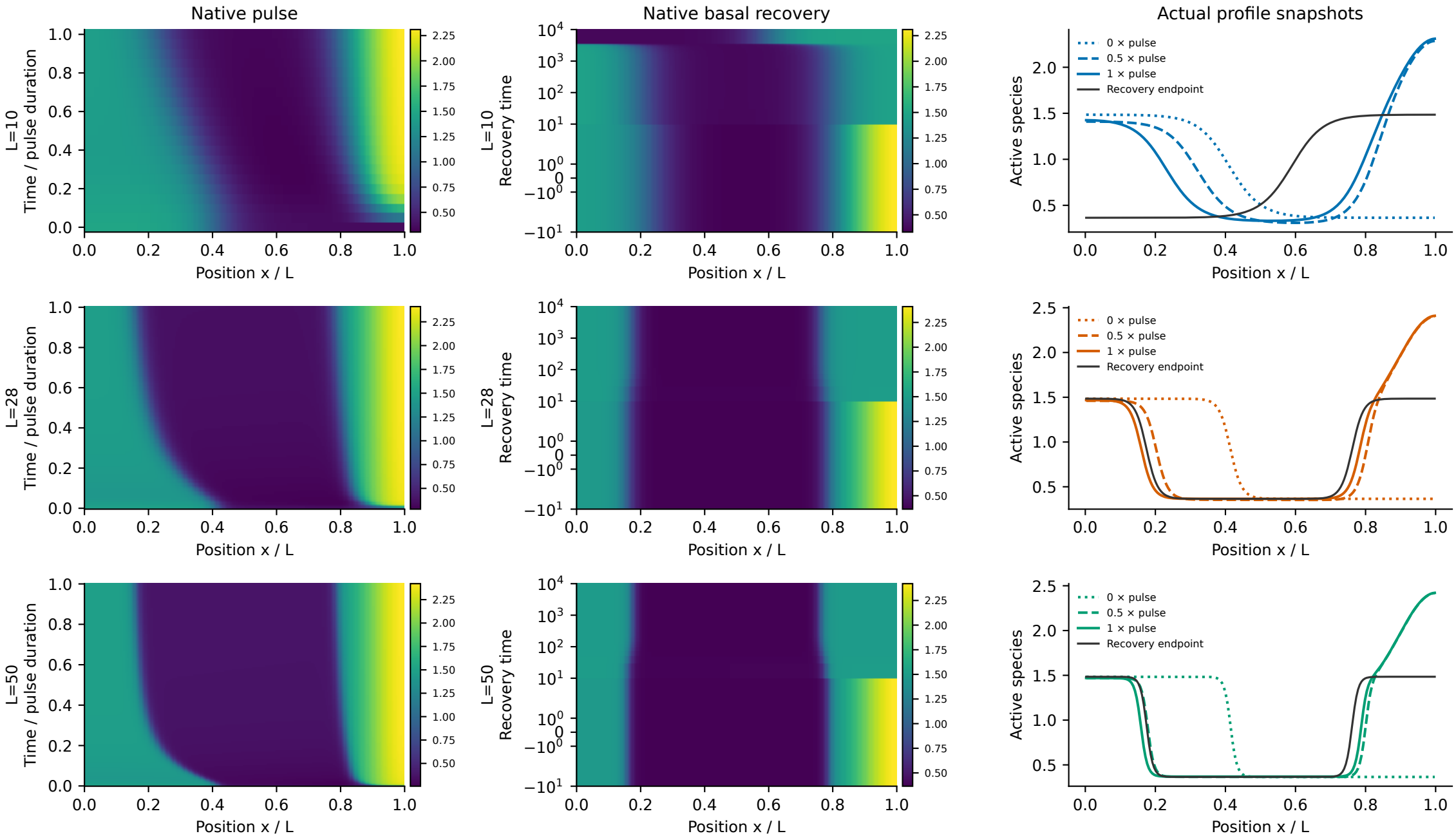








Amounts use excess above the current mid-cell median in each half; domain edges use that background plus half the initial pole contrast. Controls are fixed-dose diagnostics.


Holmes 3 · Relative width 20% · selected native spatial histories
Native pulse
Native basal recovery
Actual profile snapshots
L=10
L=28
L=50
Time / pulse duration
Recovery time
Position x / L
Active species
0 × pulse
0.5 × pulse
1 × pulse
Recovery endpoint
Native scalar-selected pulses. Separate panel color scales preserve concentration changes; these are not independently verified steady states.

Holmes 3 · Constant physical slope

Spatial coverage plus coupled Rac/Rho domain competition

Initial Rac plateau height and inactive availability are almost length independent, but the Rac domain expands. Mutual inhibition with Rho and the conserved pools supports competing spatial domains. A localized right seed need not overwrite the incumbent, even when its right activity rises quickly. The ramp peak increases with length because its physical slope is fixed. A matched-reference-peak ramp fails at the native long-cell doses, while fourfold inactive mobility preserves reversal and yields stronger recovered polarity. This supports opposing input-strength and pool/domain-cleanup contributions to the nonmonotonic curve.

Measured evidence

Old-width long/short=5.19; old-excess amount=6.067; opposite-end inactive ratio=0.9995; initial added activation-source ratio=4.997. Control failures at native doses: matched_gradient_peak 2/3, mesh_401 0/1, mobility_4x 0/3.

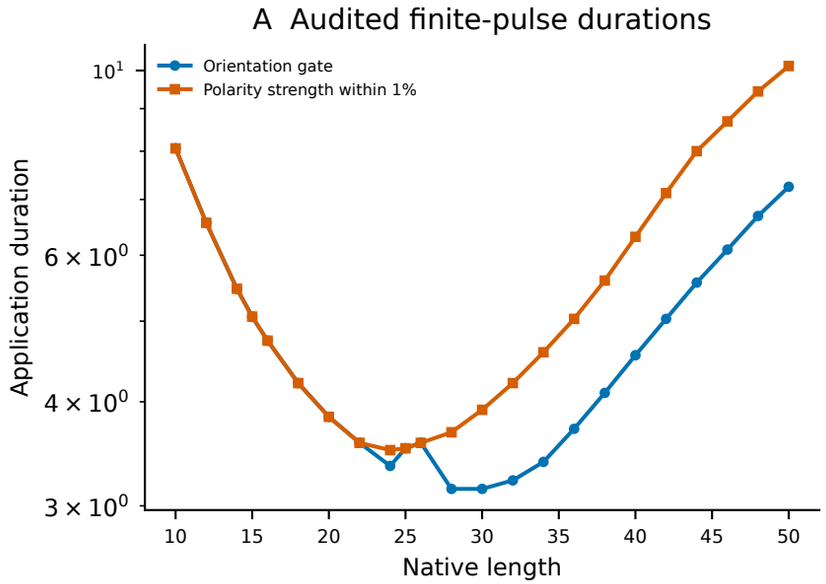
Testable prediction

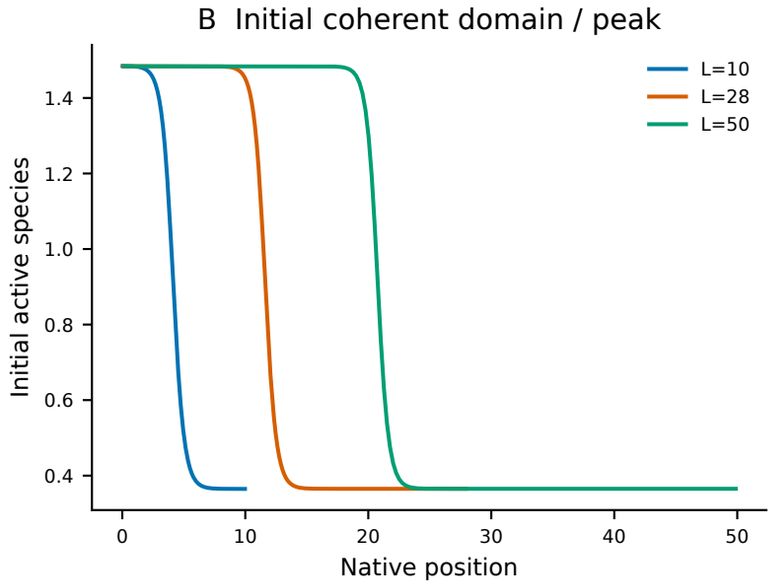
Track both Rac and Rho boundaries and inactive-pool gradients. Broader support at the same peak should rescue spatial-coverage failures; faster inactive transport can improve gradient recovery without removing every localized-patch failure.

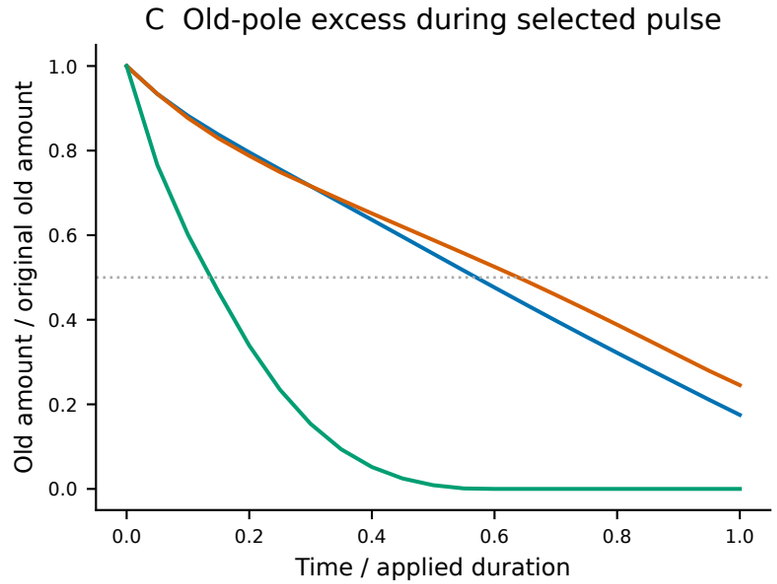
Scope of the inference

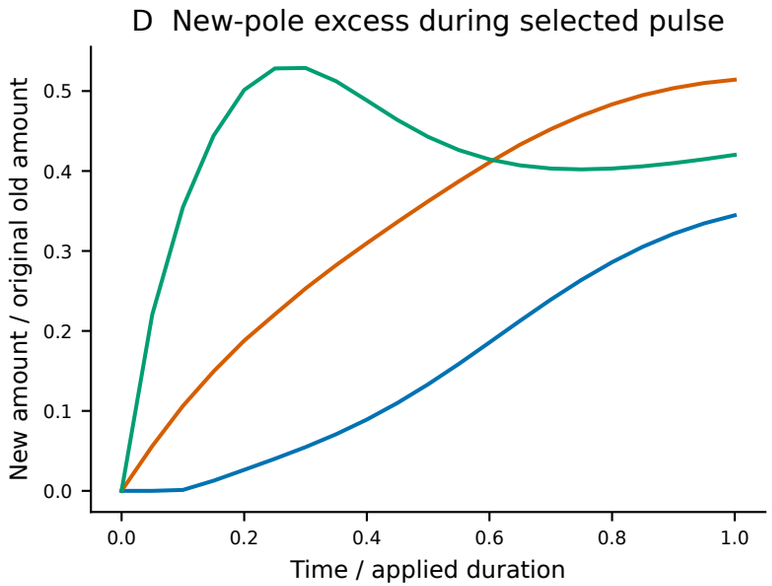
Trajectory and fixed-dose evidence in this native regime. Counterfactual thresholds were not fully searched; frozen-scale field flags and mesh sensitivity remain separate from scalar success.

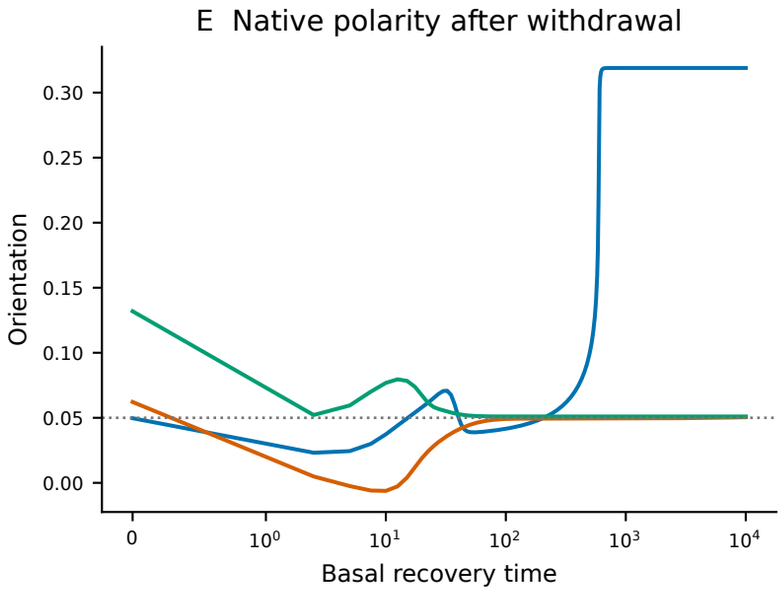
Holmes 3 · Constant physical slope

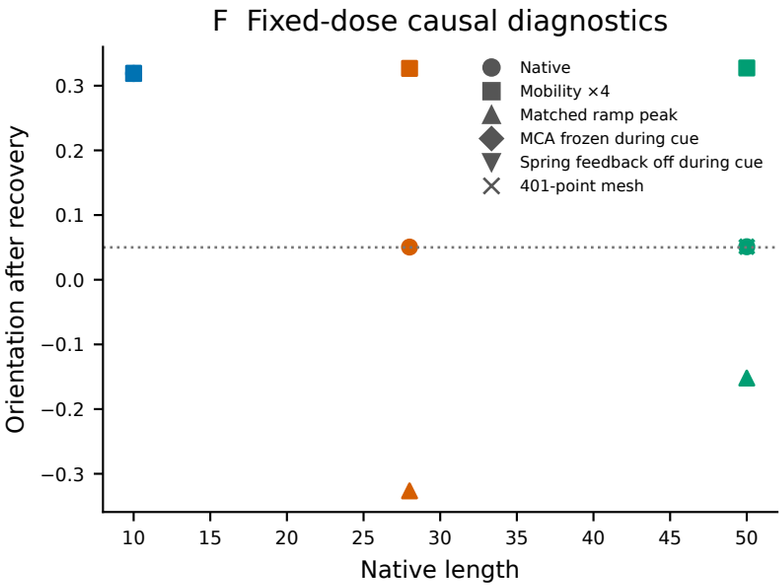




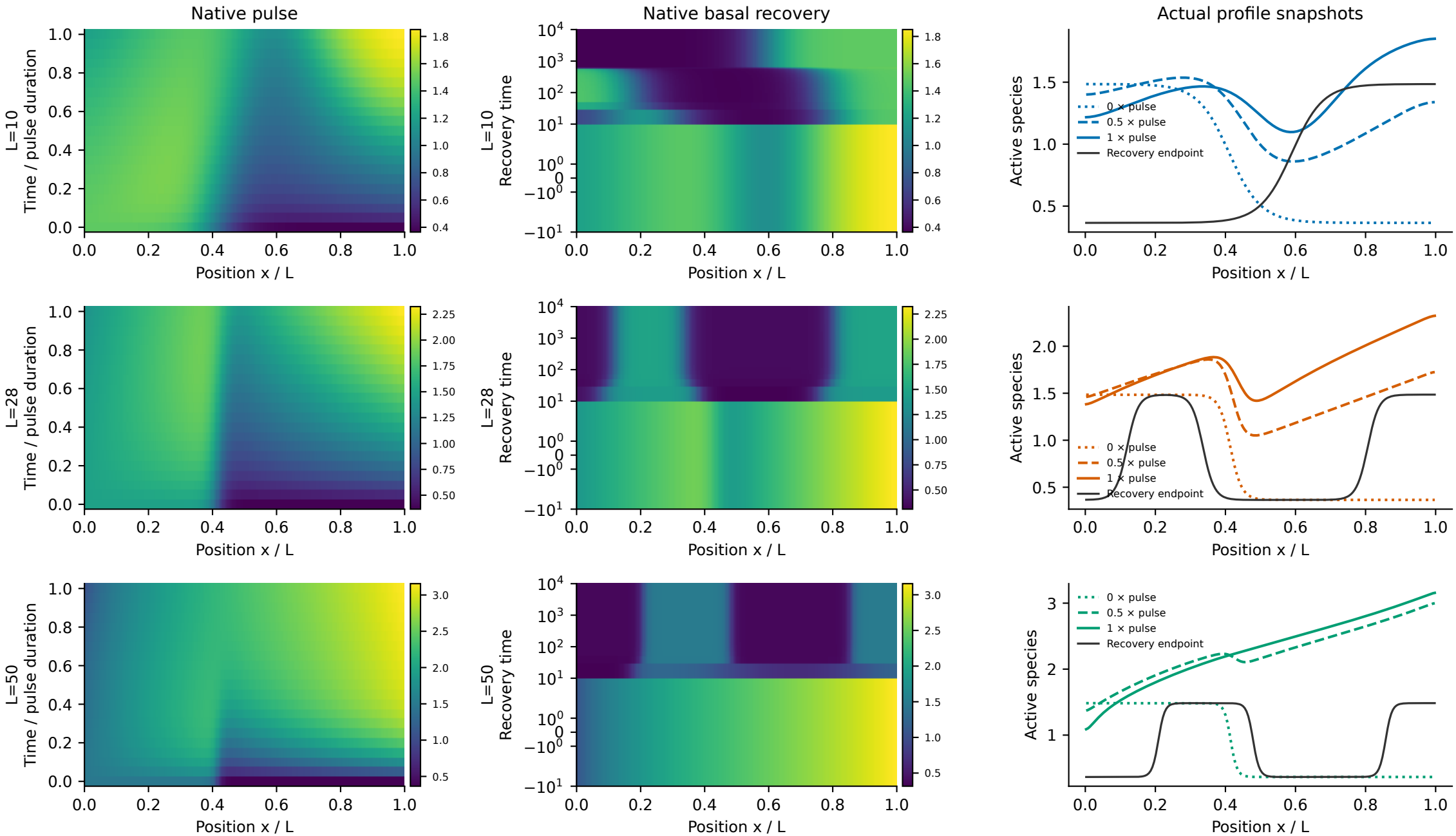








Amounts use excess above the current mid-cell median in each half; domain edges use that background plus half the initial pole contrast. Controls are fixed-dose diagnostics.



Native scalar-selected pulses. Separate panel color scales preserve concentration changes; these are not independently verified steady states.

Spring · Fixed width 3

A global extension-dependent gate coupled to polarity competition

The domain grows with length while the model links its extension to global deactivation and dilution. This is a global feedback state, distinct from deBelly adhesion memory. Cue-only removal of that feedback tests commitment at the native pulse and restores the original physical equations during recovery. The fixed support covers a decreasing fraction of a longer cell. Removing extension feedback defeats the shortest sampled cell but allows the two longer sampled cells to reverse at their native doses.

Measured evidence

Old-width long/short=5.48; old-excess amount=7.585; opposite-end inactive ratio=1.049; initial added activation-source ratio=1.049. Control failures at native doses: feedback_off_during_pulse 1/3, mesh_401 0/1, mobility_4x 0/3.

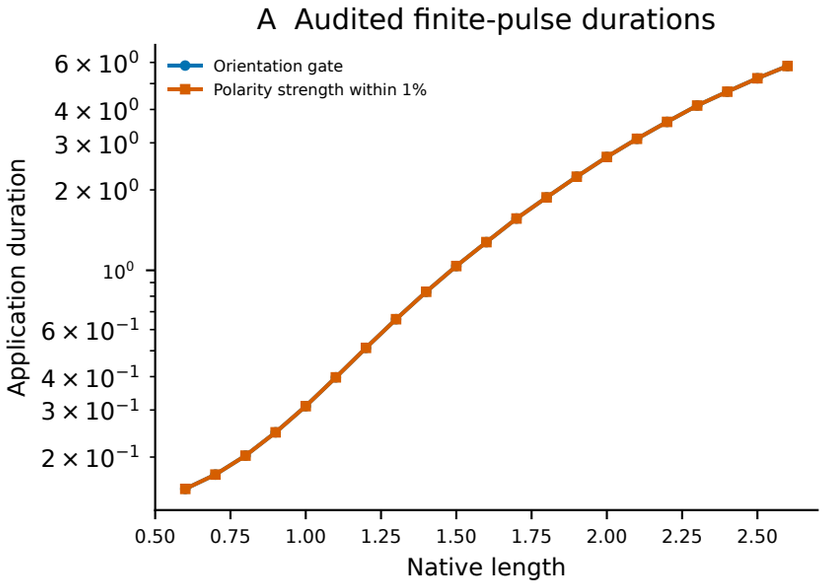
Testable prediction

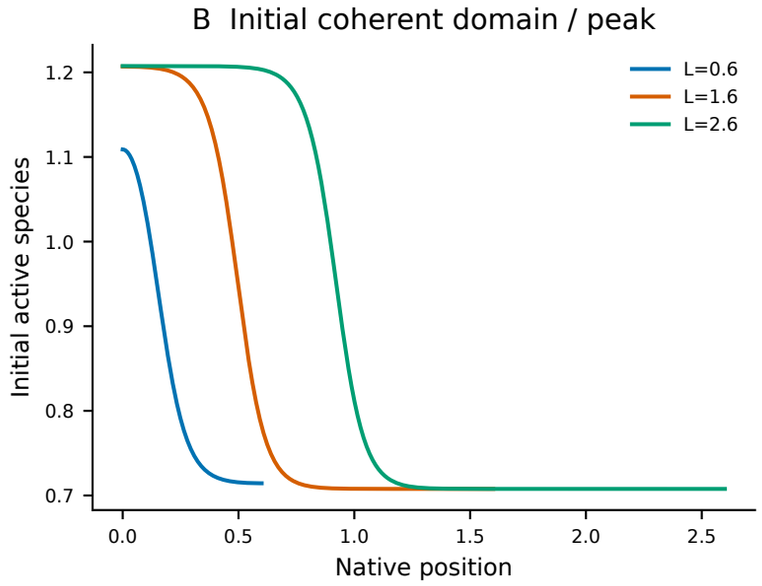
Measure physical length and extension during the cue together with Rac polarity. Changing mechanical restoring/extension feedback should shift the dose required for commitment, with effects depending on spatial cue geometry.

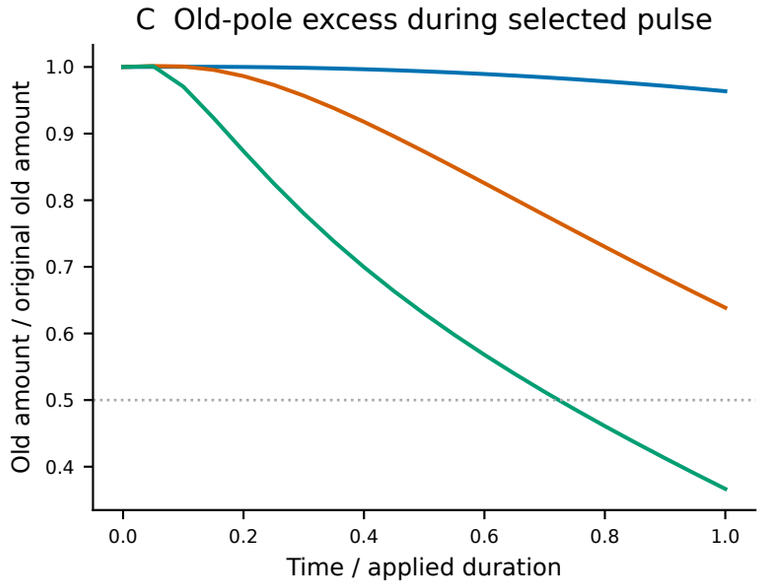
Scope of the inference

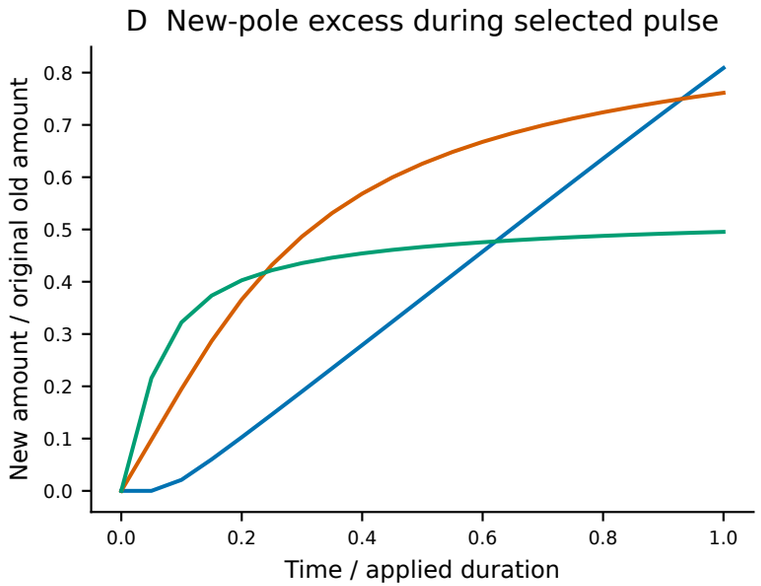
Trajectory and fixed-dose evidence in this native regime. Counterfactual thresholds were not fully searched; frozen-scale field flags and mesh sensitivity remain separate from scalar success.

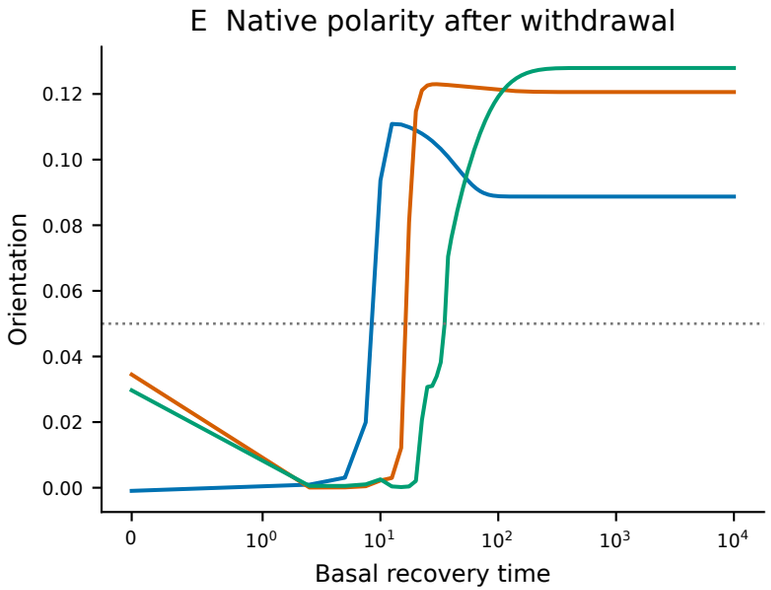
Spring · Fixed width 3

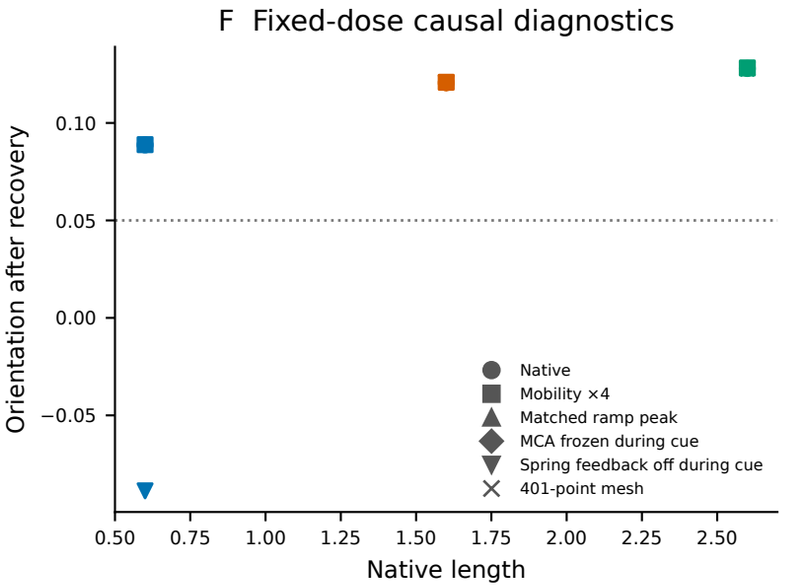




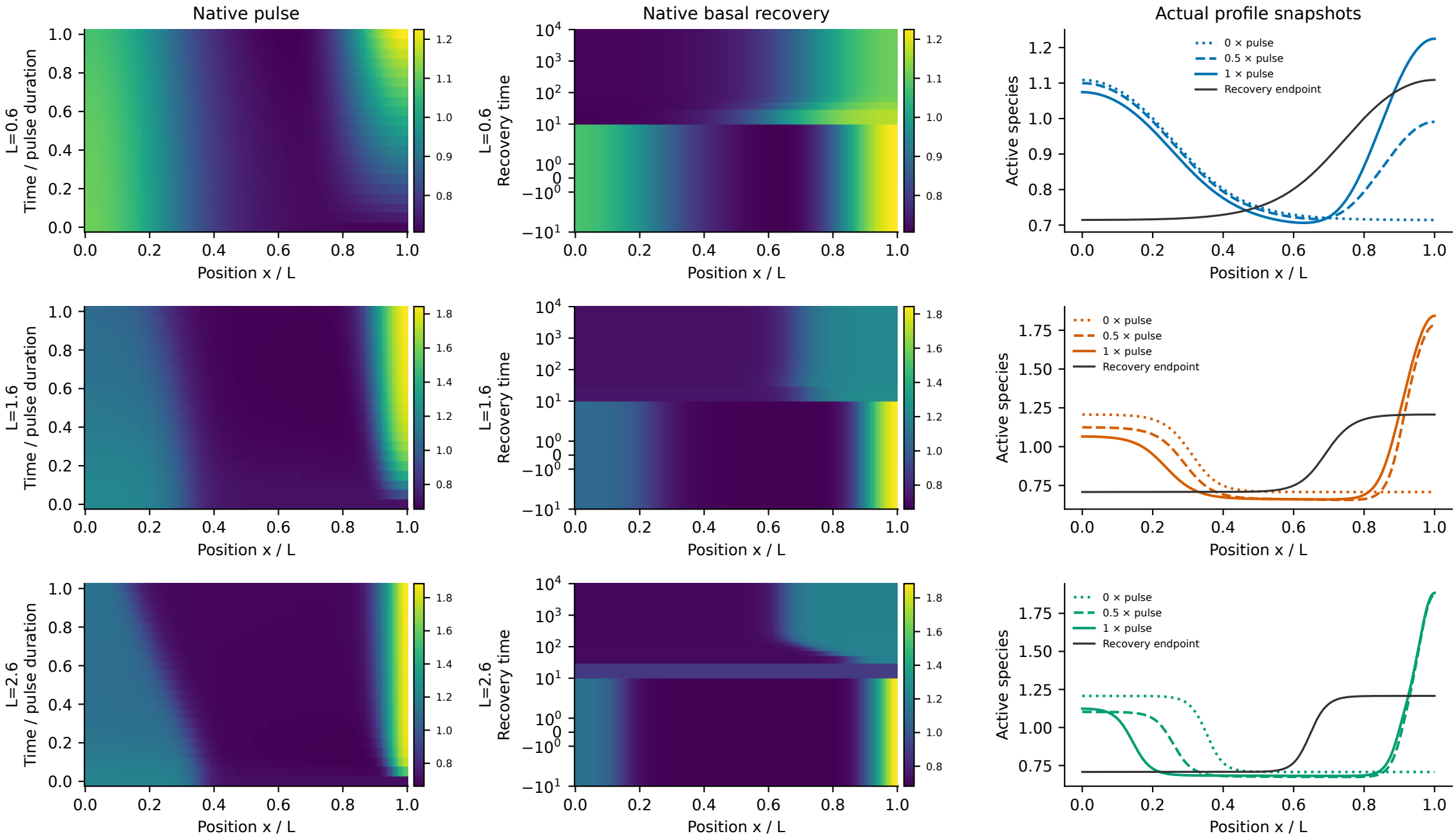




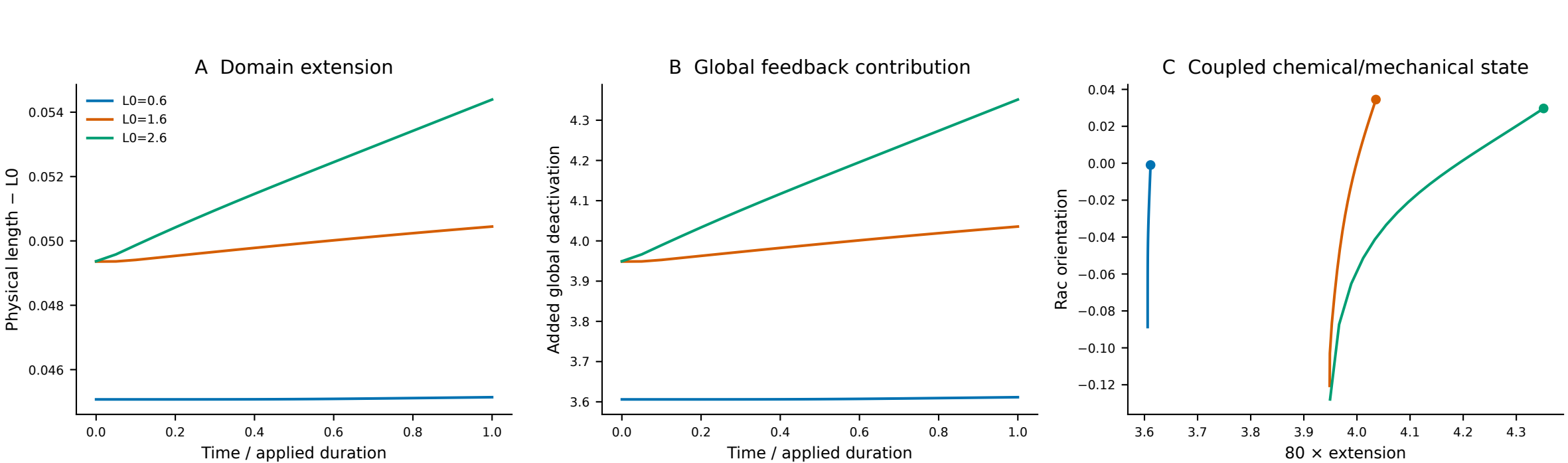




Amounts use excess above the current mid-cell median in each half; domain edges use that background plus half the initial pole contrast. Controls are fixed-dose diagnostics.



Native scalar-selected pulses. Separate panel color scales preserve concentration changes; these are not independently verified steady states.



The native deactivation rate is 3 plus 80 times extension. This global feedback is distinct from local adhesion memory. The diagnostic removes only the extension-dependent term during cue, then restores native recovery.

Spring · Relative width 20%

A global extension-dependent gate coupled to polarity competition

The domain grows with length while the model links its extension to global deactivation and dilution. This is a global feedback state, distinct from deBelly adhesion memory. Cue-only removal of that feedback tests commitment at the native pulse and restores the original physical equations during recovery. The cue keeps the same footprint fraction and peak as length grows. The duration still increases despite constant footprint fraction. Extension feedback matters at the short sampled dose but is not indispensable at the two longer sampled doses.

Measured evidence

Old-width long/short=5.48; old-excess amount=7.585; opposite-end inactive ratio=1.049; initial added activation-source ratio=1.049. Control failures at native doses: feedback_off_during_pulse 1/3, mesh_401 0/1, mobility_4x 0/3.

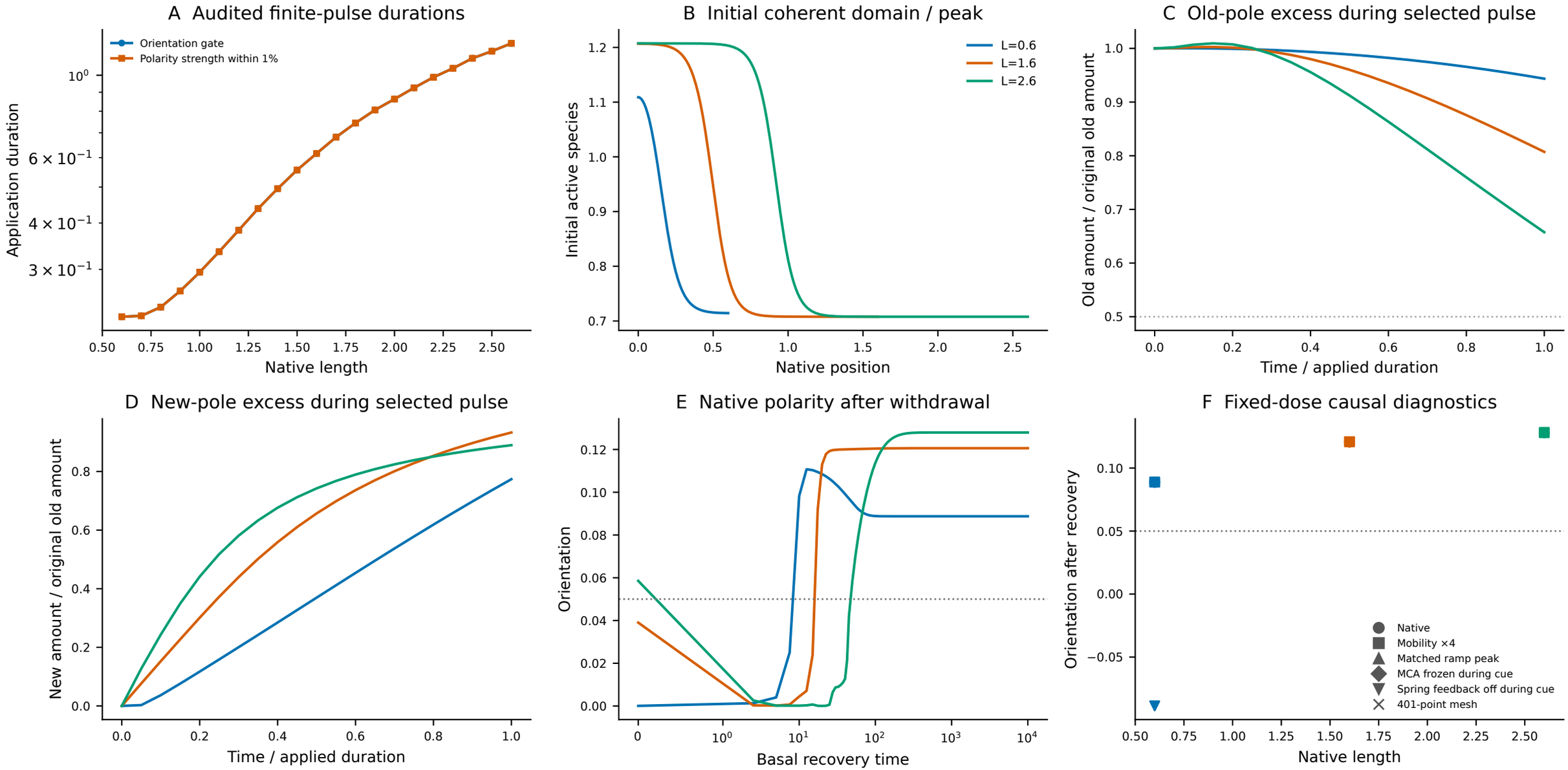
Testable prediction

Measure physical length and extension during the cue together with Rac polarity. Changing mechanical restoring/extension feedback should shift the dose required for commitment, with effects depending on spatial cue geometry.

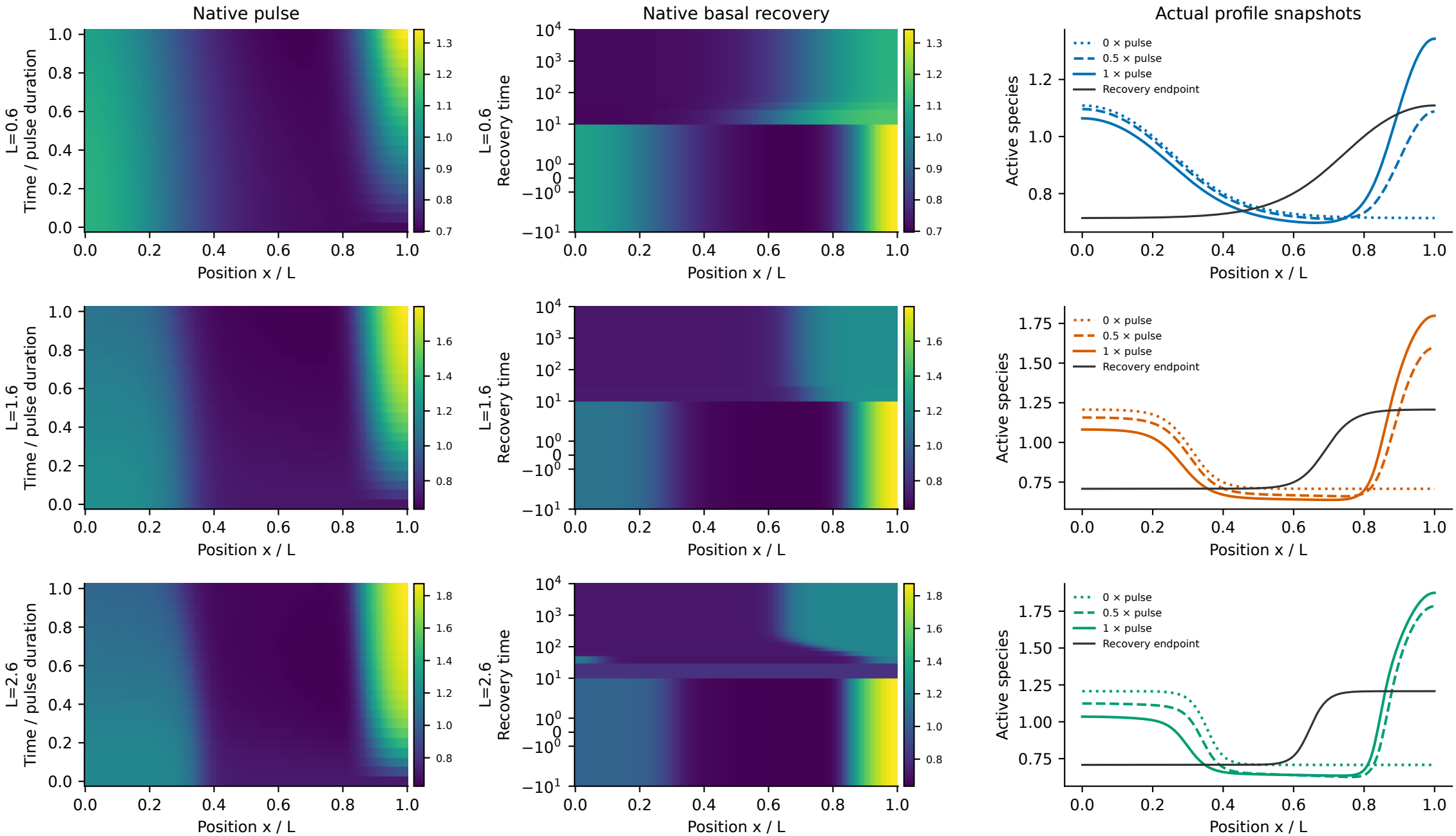
Scope of the inference

Trajectory and fixed-dose evidence in this native regime. Counterfactual thresholds were not fully searched; frozen-scale field flags and mesh sensitivity remain separate from scalar success.

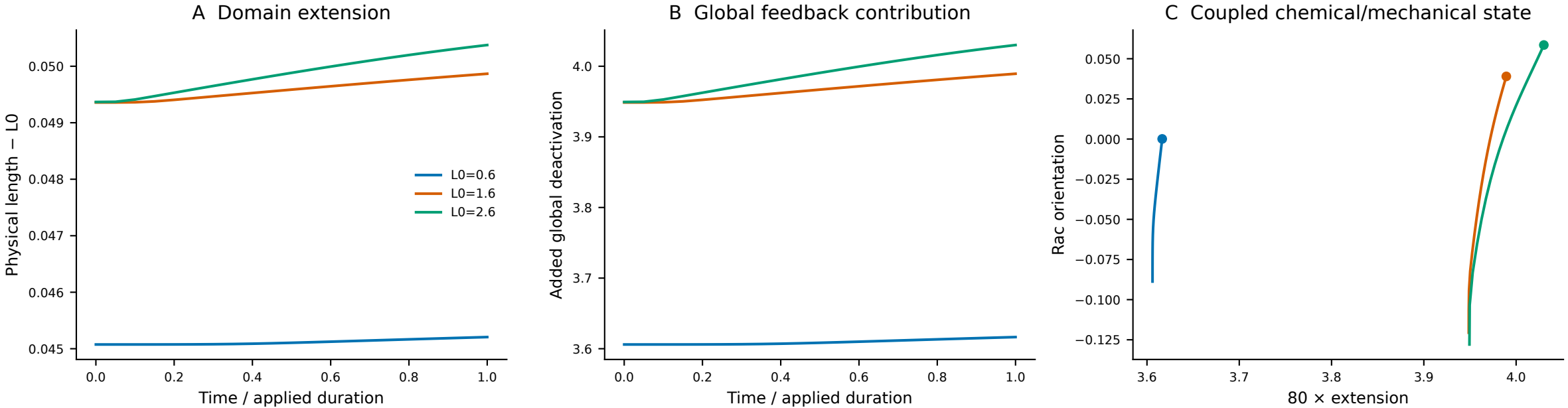
Spring · Relative width 20%



Amounts use excess above the current mid-cell median in each half; domain edges use that background plus half the initial pole contrast. Controls are fixed-dose diagnostics.



Native scalar-selected pulses. Separate panel color scales preserve concentration changes; these are not independently verified steady states.



The native deactivation rate is 3 plus 80 times extension. This global feedback is distinct from local adhesion memory. The diagnostic removes only the extension-dependent term during cue, then restores native recovery.

Spring · Constant physical slope

A global extension-dependent gate coupled to polarity competition

The domain grows with length while the model links its extension to global deactivation and dilution. This is a global feedback state, distinct from deBelly adhesion memory. Cue-only removal of that feedback tests commitment at the native pulse and restores the original physical equations during recovery. The ramp peak increases with length because its physical slope is fixed. Both matching the reference peak and removing extension feedback defeat every sampled dose above the shortest length, and feedback removal also defeats the shortest. Growing input and the global mechanical gate jointly support the flat/slightly decreasing native duration curve.

Measured evidence

Old-width long/short=5.48; old-excess amount=7.585; opposite-end inactive ratio=1.049; initial added activation-source ratio=4.308. Control failures at native doses: feedback_off_during_pulse 4/4, matched_gradient_peak 3/4, mesh_401 0/1, mobility_4x 0/4.

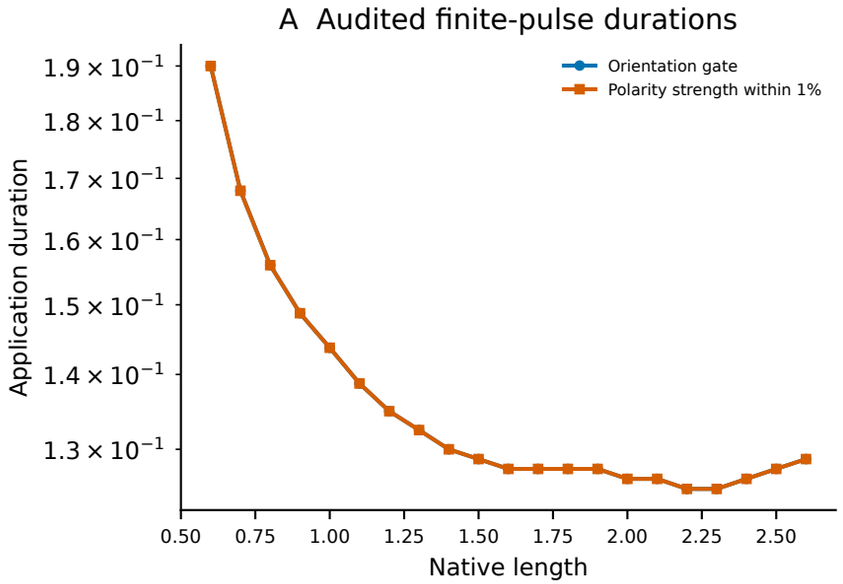
Testable prediction

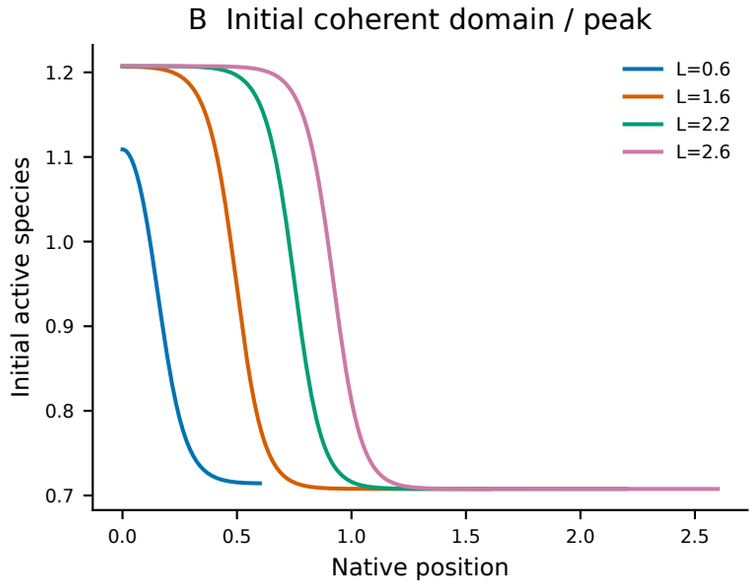
Measure physical length and extension during the cue together with Rac polarity. Changing mechanical restoring/extension feedback should shift the dose required for commitment, with effects depending on spatial cue geometry.

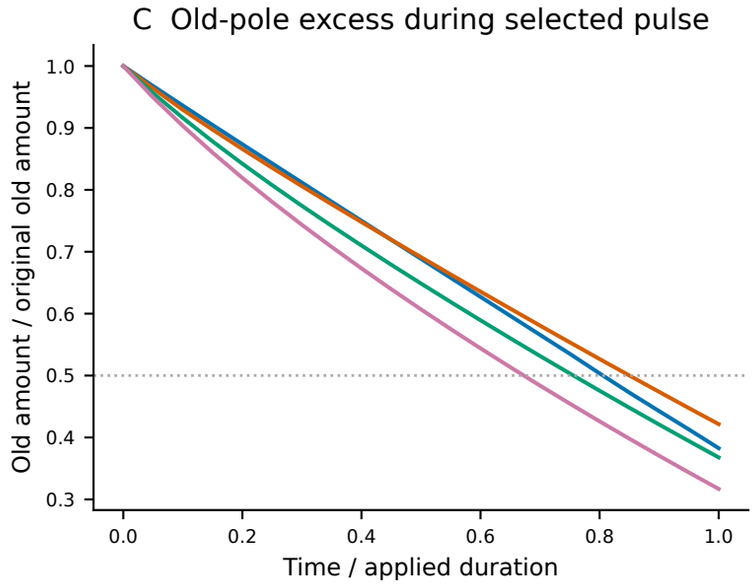
Scope of the inference

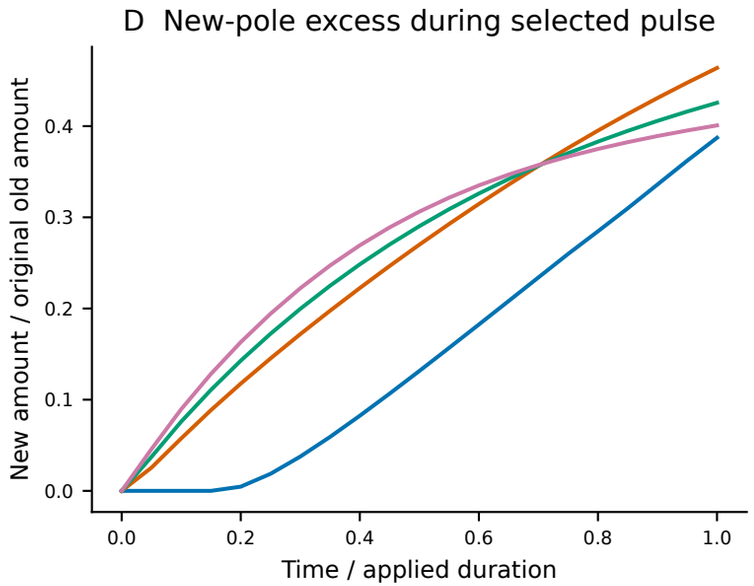
Trajectory and fixed-dose evidence in this native regime. Counterfactual thresholds were not fully searched; frozen-scale field flags and mesh sensitivity remain separate from scalar success.

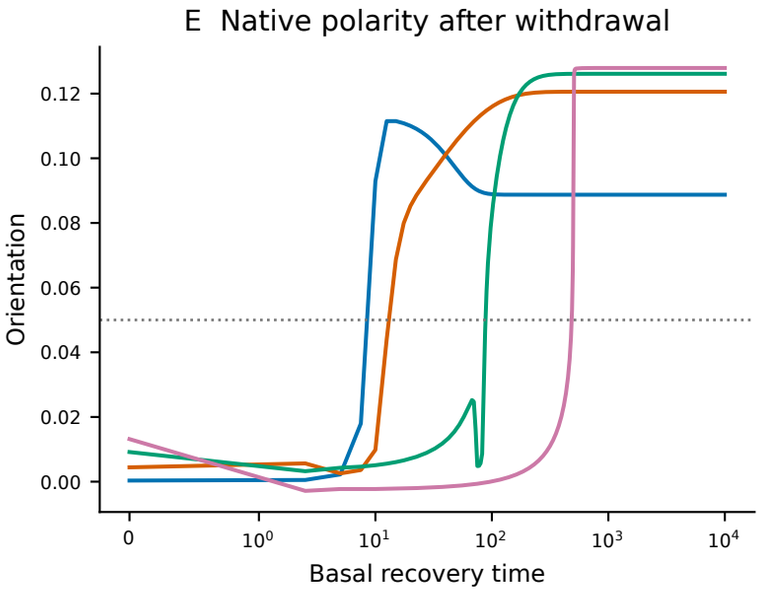
Spring · Constant physical slope

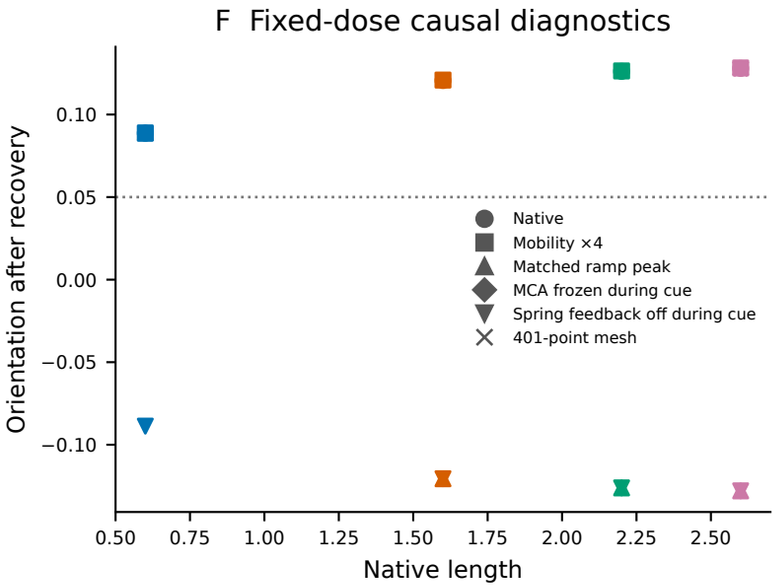




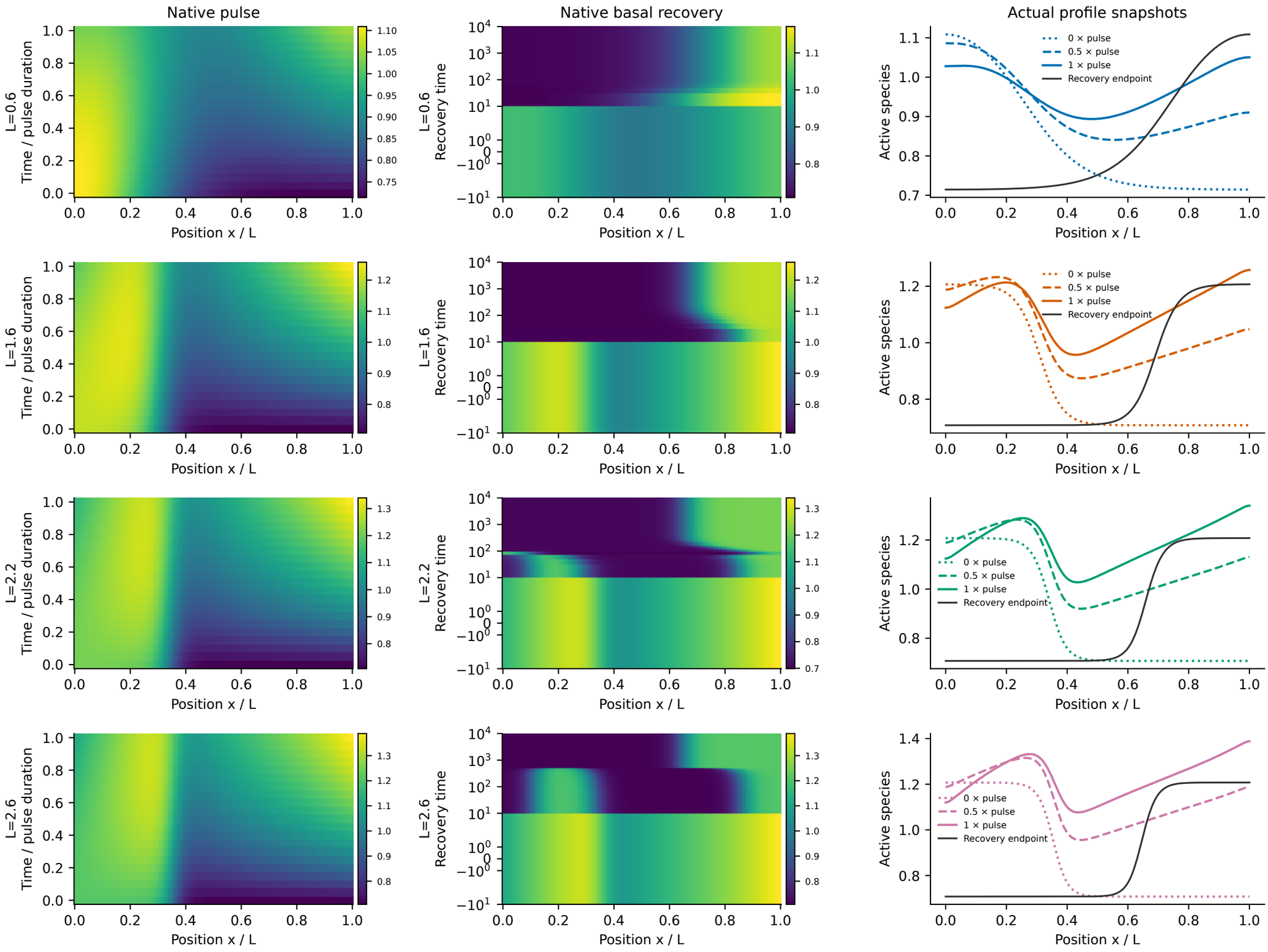




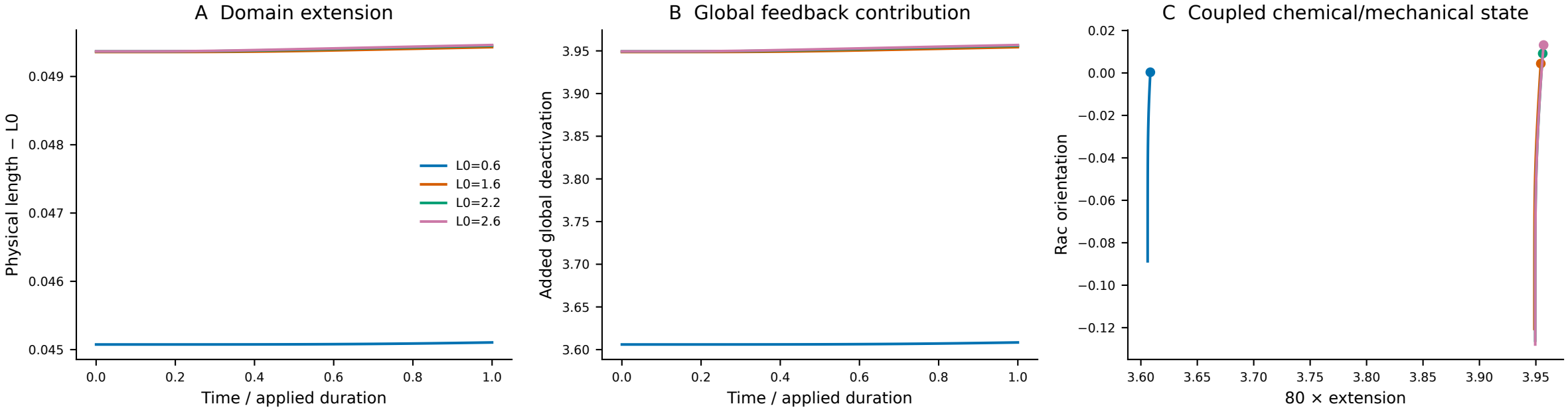




Amounts use excess above the current mid-cell median in each half; domain edges use that background plus half the initial pole contrast. Controls are fixed-dose diagnostics.



Native scalar-selected pulses. Separate panel color scales preserve concentration changes; these are not independently verified steady states.



The native deactivation rate is 3 plus 80 times extension. This global feedback is distinct from local adhesion memory. The diagnostic removes only the extension-dependent term during cue, then restores native recovery.